Indice

Capitolo 1

Introduction

1.1 Conditional expectation

1.1.1 Definitions

Definition 1.1.1 (Conditional expectation)**.** Considered:

1. a probability space: $(\Omega, \mathscr{A}, \mathbb{P})$
2. a σ-field $\mathscr{G}$ contained in $\mathscr{A}$: $\mathscr{G} \subset \mathscr{A}$ (also called sub-σ-field $\mathscr{G}$)
3. a real random variable X, which has the mean: $\mathbb{E}[|X|] < +\infty$

A conditional expectation for X given $\mathscr{G}$ is *any* random variable $V = \mathbb{E}[X|\mathscr{G}]$ such that:

1. V has mean: $\mathbb{E}[|V|] < +\infty$
2. V is $\mathscr{G}$-measurable
3. $\mathbb{E}[1_A X] = \mathbb{E}[1_A V]$, $\forall A \in \mathscr{G}$

Important remark 1 (Existence and uniqueness of conditional expectation). It can be shown that:

- at least one V satisfying these 3 properties always *exists*,
- it is almost surely *unique*: if V_1 and V_2 are both conditional expectations (satisfying the properties) they are almost surely equal:

$$\mathbb{P}(V_1 \neq V_2) = 0$$

Important remark 2 (Interpretation). We have:

- Interpretation of $\mathscr{G}$: the σ-field $\mathscr{G} \subset \mathscr{A}$ may be used to describe our *state of information*: $\forall A \in \mathscr{G}$, we already know wheter A is true or not. So $\mathscr{G}$ is collection of event which we know if they are true or not; otherwise if $A \in \mathscr{A} \setminus \mathscr{G}$ we dont' know it.

- $\mathscr{G}$-measurability of a random variable: in general a real rv is $\mathscr{A}$-measurable
$$\forall E \in \mathscr{B} = \beta(\mathbb{R}) : \exists X^{-1}(E) = \{\omega \in \Omega : X(\omega) \in E\} \in \mathscr{A}$$
Now V is $\mathscr{G}$ measurable means
$$\forall E \in \beta(\mathbb{R}) : \exists V^{-1}(E) \in \mathscr{G},$$
This is stronger condition rather than the previous because $\mathscr{G} \subset \mathscr{A}$. In general it could be that $X^{-1}(E) \in \mathscr{A}$ but $\notin \mathscr{G}$; thus the random variable would not be $\mathscr{G}$-measurable.
In this interpretation a *general random variable* Y is $\mathscr{G}$-measurable means that its value is *known* under the information $\mathscr{G}$: if I have the information $\mathscr{G}$, I know the value taken by Y. Thus the requirement that $V = \mathbb{E}[X|\mathscr{G}]$ is $\mathscr{G}$-measurable means that it's something known under the information $\mathscr{G}$
- conditional expectation as best prediction under information: we have that
  - $\mathbb{E}[X]$ is our best prediction of the random quantity X without knowing anything other information.
  - $\mathbb{E}[X|\mathscr{G}]$ on the other hand is our best prediction of X under/having the information $\mathscr{G}$ as the following thm explains

Theorem 1.1.1 (Best prediction)**.** *If the random variable has second moment ($\mathbb{E}\left[X^2\right] < \infty$), the conditional expectation V of X given $\mathscr{G}$ is the best prediction of X, in the sense that minimizes the mean squared error:*
$$\mathbb{E}\left[\left(\mathbb{E}[X|\mathscr{G}] - X\right)^2\right] = \min \mathbb{E}\left[(Z - X)^2\right] = \min \mathbb{E}\left[error^2\right] = \min MSE$$
being Z any $\mathscr{G}$-measurables rv with $\mathbb{E}\left[Z^2\right] < +\infty$.

Example 1.1.1. Suppose X is consumption of electricity in bologna tomorrow; this is a rv (since only tomorrow we will know). If X is $\mathscr{G}$-measurable, then if we have the information $\mathscr{G}$, we know the consumption. X being the consumption of electricity bologna tomorrow, $\mathscr{G}$ could be knowing that tomorrow most people of bologna go to seaside (and not many electricity is used).

Important remark 3 (Interpretation of the third requirement). If V is our best prediction of X under $\mathscr{G}$, then they cannot be independent one of each other; the third property in the definition of $V = \mathbb{E}[X|\mathscr{G}]$, $\mathbb{E}[1_A X] = \mathbb{E}[1_A V]\ \forall A \in \mathscr{G}$, is just the mathematical connection between X and V.

Remark 1 (Notation for conditioning on rvs). If Y, V, Z are any random variables then we use the notation
$$\mathbb{E}[X|Y, V, Z, \dots]$$
to mean $\mathbb{E}[X|\sigma(Y, V, Z, \dots)]$ where $\sigma(Y, V, Z, \dots)$ is the σ-field generated by Y, V, Z (least sigma field that make the random variables measurables).

Important remark 4 (Modus operandi on conditional expectation). If we our interested in $\mathbb{E}[X|\mathscr{G}]$:

1. we state that $\mathbb{E}[X|\mathscr{G}] = V$ (that can be any random variable);

2. we verify that the conjecture is true so that the three properties of conditional expectation are satisfied

Example 1.1.2. Suppose that X and Y are iid and $\mathbb{E}[|X|] < \infty$ (same for Y being iid). What can we say about $\mathbb{E}[X|X+Y]$?
eg if we know that $X + Y = 35$ what is our prediction for X? A reasonable answer is

$$\mathbb{E}[X|X+Y] = \frac{X+Y}{2}$$

This above is still a random variable, depending on the sum value at the numerator; but if we know it, it becomes known, a way to have the first component is to divide by 2, being X, Y iid.
In order to show that our guess $V = (X+Y)/2$ is actually the right answer, we have to show it satisfies the three properties of conditional expectation:

1. V do has the mean/expected value: X and Y have the mean so their means as well does;

2. V is $\mathscr{G}$-measurable: under the information $\mathscr{G} = X + Y$ its know (eg we know the value of $X+Y$ $X+Y = 35$) then we known the value of $V = 35/2$;

3. the third point is harder to show but it holds as well.

1.1.2 Properties

Remark 2. We state the main properties of $\mathbb{E}[X|\mathscr{G}]$; the first three are straightforward, the remaining less obvious.

Important remark 5. We have

1. linearity: given X, Y random variables with mean ($\mathbb{E}[|X|] < \infty, \mathbb{E}[|Y|] < \infty$), and $a, b \in \mathbb{R}$ we have

$$\mathbb{E}[aX + bY|\mathscr{G}] = a\,\mathbb{E}[X|\mathscr{G}] + b\,\mathbb{E}[Y|\mathscr{G}] \tag{1.1}$$

2. positivity: if $X \geq 0$ (non negative), then $\mathbb{E}[X|\mathscr{G}] \geq 0$

3. exp.value of constant: $\mathbb{E}[c|\mathscr{G}] = c$, with $c \in \mathbb{R}$

4. if Z is $\mathscr{G}$-measurable, then

$$\mathbb{E}[X \cdot Z|\mathscr{G}] = Z \cdot \mathbb{E}[X|\mathscr{G}]$$

   Under $\mathscr{G}$, Z is known and is treated like a constant, and goes out of expectation operator (Z is not a constant, it's a random variable, but if it's $\mathscr{G}$-measurable, behaves as a constant).
   In particular if X is $\mathscr{G}$-measurable the

$$\mathbb{E}[X|\mathscr{G}] = \mathbb{E}[X \cdot 1|\mathscr{G}] = X\,\mathbb{E}[1|\mathscr{G}] = X \cdot 1 = X$$

5. suppose $X \perp\!\!\!\perp \mathscr{G}$ meaning
$$\mathbb{P}\left(A \cap (X \in B)\right) = \mathbb{P}\left(A\right) \cdot \mathbb{P}\left(X \in B\right), \quad \forall A \in \mathscr{G}, \forall B \in \beta(\mathbb{R})$$
In that case:
$$\mathbb{E}\left[X|\mathscr{G}\right] = \mathbb{E}\left[X\right]$$
This is reasonable from logical pov: if X is independent of $\mathscr{G}$, having or not having the info of $\mathscr{G}$ is the same in order to predict X (they are independent). And thus our prediction of X under $\mathscr{G}$ coincides with our prediction without any information.

NB: non fatto quest'anno?

Example 1.1.3. An important special case where $X \perp\!\!\!\perp \mathscr{G}$ is when $\mathbb{P}\left(A\right) \in \{0,1\}, \forall A \in \mathscr{G}$ (eg if $\mathscr{G} = \{\emptyset, \Omega\}$).
We have that $X \perp\!\!\!\perp \mathscr{G}$ since $\forall A \in \mathscr{G}$ we have $\mathbb{P}\left(A\right) = 0$ or $\mathbb{P}\left(A\right) = 1$ and:
$$\begin{cases} \mathbb{P}\left(A \cap (X \in B)\right) = 0 = 0 \cdot \mathbb{P}\left(X \in B\right) = \mathbb{P}\left(A\right) \cdot \mathbb{P}\left(X \in B\right) & \text{if } \mathbb{P}\left(A\right) = 0 \\ \mathbb{P}\left(A \cap (X \in B)\right) = \mathbb{P}\left(X \in B\right) = 1 \cdot \mathbb{P}\left(X \in B\right) = \mathbb{P}\left(A\right) \cdot \mathbb{P}\left(X \in b\right) & \text{if } \mathbb{P}\left(A\right) = 1 \end{cases}$$
So if $\mathscr{G} = \{\emptyset, \Omega\}$ then $\mathbb{E}\left[X|\mathscr{G}\right] = \mathbb{E}\left[X\right]$ (here $\mathscr{G}$ have no informative value)

NB: lui in generale sembra scrivere $\subset$ intendendo $\subseteq$

6. if $\mathscr{G}_1 \subset \mathscr{G}_2 \subset \mathscr{A}$ (that is we have two sub-σ-field and we suppose they are contained, so under $\mathscr{G}_2$ we know more than $\mathscr{G}_1$) we have the so called *chain rule*:
$$\mathbb{E}\left[X|\mathscr{G}_1\right] = \mathbb{E}\left[\mathbb{E}\left[X|\mathscr{G}_2\right]|\mathscr{G}_1\right]$$
If $\mathscr{G}_1$ information is contained in $\mathscr{G}_2$ information then our prediction of X under the smaller information can be written in this way (we condition first on the bigger information and then under the smaller information).

Example 1.1.4. For instance if we know that $\mathbb{E}\left[X|\mathscr{G}_2\right] = 0$ (eg for some reason we know that with more information our prediction of X is 0), then $\mathbb{E}\left[X|\mathscr{G}_1\right]$ (our prediction on lower information)
$$\mathbb{E}\left[X|\mathscr{G}_1\right] = \mathbb{E}\left[\mathbb{E}\left[X|\mathscr{G}_2\right]|\mathscr{G}_1\right] = \mathbb{E}\left[0|\mathscr{G}_1\right] = 0$$

NB: non fatto quest'anno?

Example 1.1.5. We want to prove the equality
$$\mathbb{E}\left[X\right] = \mathbb{E}\left[\mathbb{E}\left[X|\mathscr{G}_1\right]\right]$$
is true under $X \perp\!\!\!\perp \mathscr{G}_1$, not by using $X = \mathbb{E}\left[X|\mathscr{G}_1\right]$ under independence, but by applying chain rule.
We need two σ-field one contained in the other so we take $\mathscr{G}_1 = \mathscr{G}_2 = \{\emptyset, \Omega\}$
$$\mathbb{E}\left[X\right] \overset{(1)}{=} \mathbb{E}\left[X|\mathscr{G}_1\right] \overset{(2)}{=} \mathbb{E}\left[\mathbb{E}\left[X|\mathscr{G}_2\right]|\mathscr{G}_1\right] \overset{(3)}{=} \mathbb{E}\left[\mathbb{E}\left[X|\mathscr{G}_2\right]\right]$$
where:

- (1) being $X \perp\!\!\!\perp \mathscr{G}_1$ for what we have seen in the previous point (in other words, all elements of $\mathscr{G}_1$ has prob 0 or 1 so not informative and we can condition on that)
- (2) applying the chain rule being $\mathscr{G}_1 \subset \mathscr{G}_2$
- (3) since again each element of $\mathscr{G}_1$ is prob 0 or 1 $\mathbb{E}\left[X|\mathscr{G}_2\right]$ is independent of $\mathscr{G}_1$ so we can remove the last conditioning

1.2 Conditional probability

Definition 1.2.1 (Conditional probability of an event given an event (refresher))**.** We remember that if $F, G \in \mathscr{A}$ and $\mathbb{P}(G) > 0$, then

$$\mathbb{P}(F|G) = \frac{\mathbb{P}(F \cap G)}{\mathbb{P}(G)}$$

Remark 3 (Indicator random variable (refresher)). We remember that

$$1_A = 1_A(\omega) = \begin{cases} 1 & \text{if } \omega \in A \\ 0 & \text{if } \omega \notin A \end{cases}$$
$$\mathbb{E}[1_A] = \mathbb{P}(A)$$

Remark 4. Below we *define* the general conditional probability of an event given a *sigma*-field using the indicator function

Definition 1.2.2 (Conditional probability of an event given a σ-field)**.** Given any event $F \in \mathscr{A}$ it is natural to *define* the conditional probability of F given $\mathscr{G}$ as:

$$\mathbb{P}(F|\mathscr{G}) := \mathbb{E}[1_F|\mathscr{G}]$$

Important remark 6 (Interpretation). The interpretation is the probability we give to the event F under the information $\mathscr{G}$.

Important remark 7 (Connection between the two definitions). What is the connection between $\mathbb{P}(F|G)$ (probability of F knowing that G is true) and the more general $\mathbb{P}(F|\mathscr{G}) = \mathbb{E}[1_F|\mathscr{G}]$?
Let's take a particualr field $\mathscr{G} = \{\emptyset, G, G^C, \Omega\}$ where the only information provided is wheter G is happened or not (clearly the null and samplespace events are respectively false and true).
It can be shown that, if a function $f : \Omega \to \mathbb{R}$ is $\mathscr{G}$-measurable with $\mathscr{G} = \{\emptyset, G, G^C, \Omega\}$, then can be written as linear combination of the indicators:

$$f = 1_G \cdot \alpha + 1_{G^c} \cdot \beta, \quad \alpha, \beta \in \mathbb{R}$$

Since $\mathbb{P}(F|\mathscr{G}) = \mathbb{E}[1_F|\mathscr{G}]$ must be $\mathscr{G}$-measurable (being a conditional expectation, by the second property of the definition), then it can be written as:

$$\mathbb{P}(F|\mathscr{G}) = 1_G \cdot \alpha + 1_{G^c} \cdot \beta$$

Which are α and β? To find them we use the third property of conditional expectation starting from the general case and substituting X with the indicator 1_F at the right moment

$$\begin{aligned} \forall A \in \mathscr{G}, \forall X \text{with mean:} \quad & \mathbb{E}[1_A X] = \mathbb{E}[1_A V] \\ & \mathbb{E}[1_A X] = \mathbb{E}[1_A \mathbb{E}[X|\mathscr{G}]] \\ & \mathbb{E}[1_A 1_F] = \mathbb{E}[1_A \mathbb{E}[1_F|\mathscr{G}]] \\ & \mathbb{E}[1_A 1_F] = \mathbb{E}[1_A \mathbb{P}(F|\mathscr{G})] \end{aligned}$$

where at third step we substituted X with 1_F (since can be any random variable with mean). Now, A is any event in $\mathscr{G}$, but by letting $A = G$

$$\begin{aligned} \mathbb{E}\left[1_G 1_F\right] &= \mathbb{E}\left[1_G \, \mathbb{P}\left(F|\mathscr{G}\right)\right] \\ \mathbb{E}\left[1_{F\cap G}\right] &= \mathbb{E}\left[1_G \, \mathbb{P}\left(F|\mathscr{G}\right)\right] \\ \mathbb{P}\left(F \cap G\right) &= \mathbb{E}\left[1_G \, \mathbb{P}\left(F|\mathscr{G}\right)\right] \end{aligned}$$

By substituting $\mathbb{P}\left(F|\mathscr{G}\right) = 1_G \cdot \alpha + 1_{G^c} \cdot \beta$:

$$\mathbb{P}\left(F \cap G\right) = \mathbb{E}\left[1_G(\alpha 1_G + \beta 1_{G^c})\right] = \mathbb{E}\left[\alpha 1_G + \beta \cdot 0\right] = \alpha \, \mathbb{E}\left[1_G\right] = \alpha \, \mathbb{P}\left(G\right)$$

Hence:

$$\alpha = \frac{\mathbb{P}\left(F \cap G\right)}{\mathbb{P}\left(G\right)} = \mathbb{P}\left(F|G\right)$$

Similarly by repeating the same stuff with G^c instead of G we have that

$$\beta = \mathbb{P}\left(F|G^c\right)$$

So to sum up

$$\mathbb{P}\left(F|\mathscr{G}\right) = 1_G \, \mathbb{P}\left(F|G\right) + 1_{G^c} \, \mathbb{P}\left(F|G^c\right)$$

Remark 5 (Interpretation). In the last equation if G is true ($\omega \in G$) then $\mathbb{P}\left(F|G\right)$ is our prediction, otherwise it's $\mathbb{P}\left(F|G^c\right)$. So we arrived at the abstract object $\mathbb{P}\left(F|\mathscr{G}\right)$ via calculations with elementary objects $\mathbb{P}\left(F|G\right)$ and $\mathbb{P}\left(F|G^c\right)$. Therefore the elementary definition is coherent with the general (it's a special case of the general above) in case we take $\mathscr{G} = \{\emptyset, \Omega, G, G^c\}$. We choose this special $\mathscr{G}$, because the only info provided is that G occurrs or not.

1.3 Stochastic processes

Definition 1.3.1 (Stochastic process)**.** Let $(\Omega, \mathscr{A}, \mathbb{P})$ be a probability space and T any set. A stochastic process X is any collection of random variables on $(\Omega, \mathscr{A}, \mathbb{P})$ indexed by T

$$X = \{X_t : t \in T\}$$

Important remark 8*.* Some remarks

- we will focus on real stochastic process, so called if X_t are real random variables, but omit the "real" specificatio for simplicity
- T is called *index set*: provided that can be any set, the natural interpretation is a set of times (this is not mandatory, eg it could be a set of spaces as well). if
  - T is finite or countable (eg $T = \mathbb{N}$) X is said to be a *discrete time process*[1]

[1] We have met discrete time processes: sequences of random variable are an example of discrete time processes. In case of sequences of random variable $T = \mathbb{N}$ (set of integer starting from 1 or 0).

  - T is an interval (eg usually $[0, +\infty)$ or $[0, 1)$), X is called a *continuous time process.*

- For each $t \in T$ we have a random variable $X_t : \Omega to \mathbb{R}$ which is measurable (meaning $\forall B \in \beta(\mathbb{R}) : \exists X_t^{-1}(B) = A \in \mathscr{A}$)

  Example 1.3.1. X_t could be the consumption of electricity in Bologna at time t.

- A stochastic process is actually a function of two variables $t \in T$ and $\omega \in \Omega$: infact for fixed $t \in T$, X_t is a real random variable thus a function $\Omega \to \mathbb{R}$. So we can write

$$X_t(\omega) \quad \text{or} \quad X(t, \omega)$$

Definition 1.3.2 (Path of a stochastic process X)**.** If we fix an $\omega \in \Omega$, the deterministic function $p : T \to X(t, \omega)$ is called the path (or trajectory) of X for ω.

Remark 6. How many path are there? we have a path for each point $\omega \in \Omega$ (and are depicted as a single time series)

Definition 1.3.3 (Filtration)**.** If $T \subset \mathbb{R}$, a filtration is a *nested* collection of sub-σ-fields of $\mathscr{A}$, indexed by T. That is

- is a sequence $(\mathscr{F}_t : t \in T)$ where each $\mathscr{F}_t \subset \mathscr{A}$ is a sub-σ-field
- which is *nested*, meaning $\mathscr{F}_s \subset \mathscr{F}_t$ if $s < t$

Example 1.3.2. For us the most important case, if $T = \{0, 1, 2, \ldots\}$ a filtration is $\mathscr{F}_0 \subset \mathscr{F}_1 \subset \mathscr{F}_2 \subset \ldots \subset \mathscr{A}$

Important remark 9 (Interpretation). $\mathscr{F}_0$ is our information at time 0, $\mathscr{F}_1$ is our informatin at time 1. With a filtration, we assume that as time goes on information increase so $\mathscr{F}_t \subset \mathscr{F}_{t+1}$ (tomorrow we have all the information of today plus maybe something else).

Definition 1.3.4 (Process adapted to a filtration)**.** A process X is said to be adapted to a filtration $\mathscr{F}_t$ if X_t is $\mathscr{F}_t$-measurable $\forall t \in T$ (that is $X_t^{-1}(B) \in \mathscr{F}_t, \forall B \in \beta\mathbb{R}$).

Remark 7 (Interpretation). If a single random variable X is $\mathscr{G}$-measurable, then under the information $\mathscr{G}$ we know the value taken by X.
In a random process X is adapted to a filtration $\mathscr{F}_t$, at time t we know the value taken by $X_0, X_1, \ldots, X_t$ (and maybe other things)

Example 1.3.3 (σ-field generated by the process X until t)**.** Suppose the filtration $\mathscr{F}_t^* = \sigma(X_0, X_1, \ldots, X_t)$ be the σ-field generated by the process X until n (that is the least σ-field which makes $X_0, X_1, \ldots, X_t$ measurable).
Is the process X adapted to the filtration $\mathscr{F}_t^*$? Yes, by definition; if we are in $\mathscr{F}_t^*$ the process is necessarily adapted.

Definition 1.3.5 (Stopping time for a filtration)**.** A stopping time for the filtration $\mathscr{F}_0 \subset \mathscr{F}_1 \subset \ldots \subset \mathscr{A}$ is any function $\mathrm{T} : \Omega \to \{+\infty, 0, 1, \ldots\}$ such that the event $\{\tau = T(\omega) = n\} \in \mathscr{F}_n$, $\forall n \geq 0$.

Important remark 10 (Interpretation). τ

- is a random time (look at the function)
- the first time "something" we are interested in happens
- could be $\tau = +\infty$ if the event we are waiting for never happens

The condition that the event $\{\tau = n\} \in \mathscr{F}_n \forall n \geq 0$ means that the event $\{\tau = n\}$ is True or False (known) under the information $\mathscr{F}_n$.

Example 1.3.4 (Stopping time important example)**.** Let:

- $\mathscr{F}_1 \subset \mathscr{F}_2 \subset \ldots \subset \mathscr{A}$ be a filtration
- $X_0, X_1, \ldots$ a sequence of random variable adapted to $(\mathscr{F}_n)$
- let $A \in \beta(\mathbb{R})$ be an event of interest, we're interested in entering in the set A
- $\tau = \inf\{n \geq 0 : X_n \in A\}$ (τ is the first time the event A happen) with the convention $\inf\{\emptyset\} = +\infty$. There are 2 possible situations:
  - $X_n \notin A, \forall n \geq 0$: in this case the set $\{n \geq 0 : X_n \in A\}$ is empty, so $\tau(\emptyset) = +\infty$ by convention;
  - $X_n \in A$ for some n: in this case τ is the first n such that $X_n \in A$

If X_n is price of actions at time n and $A = [1, +\infty)$ represent the decision to sell the action (when the price become greater or equal to 1), then

$$\tau = \text{first time since that the price is} \geq 1.$$

We need to set $\inf(\emptyset) = +\infty$ because it can be that never the price of the action becomes above 1.

Example 1.3.5. If the stopping time is defined as $\tau = \inf\{t \geq 0 : X_t = a\}$ with $a \in \mathbb{R}$ it's the first time such that $X = a$

NB: non fatto quest'anno

Definition 1.3.6 (Finite dimensional distributions of a process X)**.** Given a process X, these are all the distributions of the n-variate random vector $(X_{t_1}, \ldots, X_{t_n})$, $\forall n \geq 1$, $\forall t_1, \ldots, t_n \in T$.

Important remark 11. What happens in real problems is that one first choose the finite dimensional distributions he/she wants and then looks for a process X having such finite dimensional distributions.

Remark 8. This is not always in general possible.
For instance suppose that we want a process where the bivariate random variables are normal while the univariate random variables are binomial. That is where $(X_{t_1}, X_{t_2}) \sim N$, $\forall t_1, t_2 \in \mathbb{N}$ and X_t is binomial $\forall t$.
This is not possible since if the n-variate (eg bivariate) is normal all the marginals (eg univariate) are still normal. So here can't a marginal being binomial.

Remark 9. However there exists some theorems (the most important of which is the Kolmogorov consistency theorem) which give conditions on the finite dimensional distribution that guarantee the existence of a process X having such finite dimensional distributions.

Remark 10. So far we've talked about stochastic processes in general, next we start by looking at several stochastic processes, starting from a special/important type, the martingale.

Capitolo 2

Martingales

2.1 Introduction

Remark 11. Martingales[1] are a type of stochastic process that represents a sequence of random variables in which, given the current value of the process and all the information available up to that point, the expected value of future variables is equal to the current value. In other words, a martingale describes a situation where there is no predictable trend: what happens in the future does not depend on what has happened in the past, but only on the present value.

Definition 2.1.1 (Martingale)**.** Let $(\mathscr{F}_n)$ be a filtration and (X_n) a sequence of real random variable; then (X_n) is a martingale with respect to $(\mathscr{F}_n)$ if:

1. $\mathbb{E}\left[|X_n|\right] < +\infty$, $\forall n$ (all random variables have the mean)
2. (X_n) is adapted to $(\mathscr{F}_n)$
3. $\mathbb{E}\left[X_{n+1}|\mathscr{F}_n\right] = X_n$, $\forall n$ (*qualifying* condition)

Example 2.1.1. Suppose X_n is the price of an action at time n: if the process is a martingale our prediction of price at time $n+1$, given all we know today, is equal to the price today.
This is an *assumption* we make on the process: sometimes is reasonable, in other situations not.

Important remark 12 (Martingales as fair games). If at the casino at time n we have X_n euros $X_{n+1} - X_n$ is the winning at time $n+1$ (could be negative as well).
If (X_n) is a martingale the expected win conditional to all the information we have at time n is 0 (*fair game*):

$$\mathbb{E}\left[X_{n+1} - X_n|\mathscr{F}_n\right] \overset{(1)}{=} \mathbb{E}\left[X_{n+1}|\mathscr{F}_n\right] - \mathbb{E}\left[X_n|\mathscr{F}_n\right] \overset{(2)}{=} X_n - X_n = 0$$

where:

- (1) by property of conditional expectation

[1] The name martingale ("martingala" in italian) comes from betting: (some types of bettings are called martingale)

- (2) by definition of martingale

Thus the martingale can be interpreted as a sequence of fair games.

Remark 12. If the process is a martingale is either a submartingale or supermartingale (but not necessary the contrary, a process could be a submartingale but not a martingale)

Definition 2.1.2 (Submartingale)**.** We have that $\mathbb{E}[X_{n+1}|\mathscr{F}_n] \geq X_n$

Definition 2.1.3 (Supermartingale)**.** We have that $\mathbb{E}[X_{n+1}|\mathscr{F}_n] \leq X_n$

Proposition 2.1.1 (Martingale and rvs expected value)**.** *If (X_n) is a martingale then all the rvs involved have the* same *mean:*

$$\mathbb{E}[X_n] = \mathbb{E}[X_0], \forall n$$

Dimostrazione. In fact

$$\mathbb{E}[X_{n+1}] = \mathbb{E}[\mathbb{E}[X_{n+1}|\mathscr{F}_n]] = \mathbb{E}[X_n]$$

So:

- for $n = 0$ we get $\mathbb{E}[X_1] = \mathbb{E}[X_0]$
- for $n = 1$ we get $\mathbb{E}[X_2] = \mathbb{E}[X_1] = \mathbb{E}[X_0]$
- thus by induction $\mathbb{E}[X_n] = \mathbb{E}[X_0]$, $\forall n$

□

Proposition 2.1.2. *If (X_n) is a martingale with respect to some filtration $(\mathscr{F}_n)$, then (X_n) is also a martingale with respect to $(\mathscr{F}_n^*) = \sigma(X_0, X_1, \dots, X_n)$ (σ-field generated by the process X until n, called* canonical filtration*).*[2]

Dimostrazione. To prove it, we note that $\mathscr{F}_n^* \subset \mathscr{F}_n$:

- if X_n is a martingale with respect to $\mathscr{F}_n$, then by the second condition it's adapted to $\mathscr{F}_n$ and thus $\mathscr{F}_n^* \subset \mathscr{F}_n$
- in other words at time n we have at least/surely the information provided by the process up to that time, plus maybe something else

Thus one can use the chain rule for conditional expectation and show that third condition of the definition holds for $\mathscr{F}_n^*$:

$$\mathbb{E}[X_{n+1}|\mathscr{F}_n^*] \overset{(1)}{=} \mathbb{E}[\mathbb{E}[X_{n+1}|\mathscr{F}_n]|\mathscr{F}_n^*] \overset{(2)}{=} \mathbb{E}[X_n|\mathscr{F}_n^*] \overset{(3)}{=} X_n$$

where:

- in (1) we use the chain rule.
- in (2) we substitute $\mathbb{E}[X_{n+1}|\mathscr{F}_n] = X_n$ by the martingale main property.

[2] La proposizione si trova anche scritta come

$$\mathbb{E}[X_{n+1}|X_1, \dots X_n] = \mathbb{E}[\mathbb{E}[X_{n+1}|\mathscr{F}_n]|X_1, \dots X_n] = \mathbb{E}[X_n|X_1, \dots X_n] = X_n$$

- in (3) given that X_n is adapted to $\mathscr{F}_n^*$ (so info of X_n is included in $\mathscr{F}_n^*$)

$\square$

Remark 13. Thus any time we know that X_n is a martingale with respect to any filtration we know it's also with respect to the "canonical" filtration ($\mathscr{F}_n^*$) derived by the process up to time n.[3]

2.2 Random walk

Remark 14. The most important random process and sometime example of martingale, is the random walk.

Definition 2.2.1 (Random walk)**.** Any sequence (X_n) is a random walks if can be written as $X_n = X_0 + \sum_{i=1}^{n} Z_i$ where:

1. (Z_i) is iid sequence
2. X_0 is independent of (Z_i).

Remark 15 (Interpretation). In a sense, we are making a walk on the real axis, at every step we start from the previous position and modify it by an independent value.

Example 2.2.1 (Symmetric random walk on $\mathbb{Z} = \{\ldots, -1, 0, 1, \ldots\}$)**.** Important special case of random walk where:

- X_0 in $\mathbb{Z}$ (take an integer at time 0 randomly)
- Z_i are such that

$$\mathbb{P}(Z_i = 1) = \mathbb{P}(Z_i = -1) = \frac{1}{2}$$

At each step we go left or right of 1 with same probability.

Remark 16. A random walk is not necessarily a martingale: the thm below gives conditions under which this happens.

Theorem 2.2.1 (Randow walk and martingale)**.** *Let $\mathscr{F}_n = \sigma(X_0, X_1, \ldots, X_n) = \sigma(X_0, Z_1, \ldots Z_n)$; the random walk (X_n) is a martingale $\iff$:*

1. *X_0, Z_i have mean ($\mathbb{E}[|X_0|] < \infty$, $\mathbb{E}[|Z_1|] < \infty$) and*
2. *furthermore Z_i has mean 0 ($\mathbb{E}[Z_1] = 0$, as the others, being iid)*

NB: Exam questions: 1) what is a martingale? 2) what is a natural example of martingale? a random walk satysfying this thm.

Dimostrazione. Solo di $\Leftarrow$:

$$\mathbb{E}[X_{n+1}|\mathscr{F}_n] = \mathbb{E}[X_n + Z_{n+1}|\mathscr{F}_n] = X_n + \mathbb{E}[Z_{n+1}|\mathscr{F}_n] \overset{(1)}{=} X_n + \mathbb{E}[Z_{n+1}] = X_n$$

where in (1) because of $Z_{n+1} \perp\!\!\!\perp \mathscr{F}_n$ and in (2) by $\mathbb{E}[Z_i] = 0$ $\square$

Example 2.2.2. Is symmetric random walk on $\mathbb{Z}$ a martingale? Yes if X_0 has the mean, because Z_1 has mean and it's 0.

[3] Eg if while reading a book the author does not specify which filtration a X_n martingale is adapted to, then we can suppose/assume the filtration is this one ($\mathscr{F}_n^*$).
Suppose we know (X_n) is a martingale but no one specify which is the filtration; we are always allowed to take $\mathscr{F}_n^*$ as filtration.

2.3 Martingales and stopping time

Remark 17 (Connection martingale and stopping time). There is a nice connection between martingales and stopping time; some notation/definition before.

Important remark 13 (Notation). Given $a, b \in \mathbb{R}$, $a \wedge b$ means $\min(a, b)$.

Definition 2.3.1 (Stopped sequence)**.** Given a sequence of rvs (X_n) and a stopping time τ it's defined as:

$$(X_{n\wedge\tau}) = X_0, X_1, \dots, X_\tau, X_\tau, X_\tau, \dots$$

Remark 18. So the sequence is like the original up to τ and then all the following elements are X_τ.

Theorem 2.3.1 (Martingale and stopping time)**.** *If (X_n) is a martingale and τ is a stopping time, then the stopped sequence $(X_{n\wedge\tau})$ is still a martingale.*

Example 2.3.1 (Example of martingale: Gambler's ruin problem)**.** Player A (me, with $a > 0$ euros at beginning) and B (with $b > 0$ euros) toss a coin multiple times: if it comes tails B gives 1 euro to A, while with head the converse. The winner is the one that ruins the other (its euros going to zero).
There are 3 outcomes at least in principle: player A wins, player B wins, or the game never ends (actually it can be shown that this situation happens with probability 0).
Let's model this as a symmetric random walk on integers (with 0 starting point): X_n be my total *winning* (obtained due due to game only) at time n

$$X_n = X_0 + \sum_{i=1}^{n} Z_i$$

with $X_0 = 0$ (winnning at time 0), Z_i iid with $\mathbb{P}(Z_i = 1) = \mathbb{P}(Z_i = -1) = \frac{1}{2}$.
I win/end game if $X_n = b$ (i ruin player B) happens before $X_n = -a$ (i am ruined). Time when the game ends is the first occurrence:

$$\tau = \inf \{n \geq 0 : X_n = b \text{ or } X_n = -a\}$$

Let X_τ be my winnings at the end of the game: it's a discrete rv. Thus expected final winning $\mathbb{E}[X_\tau]$ is:

$$\begin{aligned}
\mathbb{E}[X_\tau] &= b\,\mathbb{P}(X_\tau = b) + (-a)\,\mathbb{P}(X_\tau = -a) \\
&= b\,\mathbb{P}(X_\tau = b) + (-a)(1 - \mathbb{P}(X_\tau = b)) \\
&= b\,\mathbb{P}(X_\tau = b) - a + a\,\mathbb{P}(X_\tau = b)
\end{aligned}$$

Thus

$$\mathbb{P}(X_\tau = b) = \mathbb{P}(\text{A wins}) = \frac{\mathbb{E}[X_\tau] + a}{a + b}$$

Now we want to evaluate $\mathbb{E}[X_\tau]$: since the situation 3 (game never ends) has probability 0, the games ends with probability 1 so τ is finite almost surely ($\mathbb{P}(\tau < +\infty) = 1$). Thus being finite:

$$X_\tau = \lim_{n\to+\infty} X_{\tau\wedge n}$$

hence,

$$\mathbb{E}[X_\tau] = \mathbb{E}\left[\lim_{n\to+\infty} X_{\tau\wedge n}\right] \overset{(1)}{=} \lim_{n\to+\infty} \mathbb{E}[X_{\tau\wedge n}]$$
$$\overset{(2)}{=} \lim_{n\to+\infty} \mathbb{E}[X_{\tau\wedge 0}] = \lim_{n\to+\infty} \mathbb{E}[X_0] = \mathbb{E}[X_0] = 0$$

where in:

- (1) lim goes out of expectation (we can interchange them here, take it as given);
- (2) since X_n is a martingale (being a symmetric random walk on the integers) then the stopped sequence $X_{\tau\wedge n}$ is a martingale as well; and being a martingale has constant mean value ($\mathbb{E}[X_{\tau\wedge n}] = \mathbb{E}[X_{\tau\wedge 0}]$)

So finally:

$$\mathbb{P}(\text{I win}) = \frac{\mathbb{E}[X_\tau] + a}{a+b} = \frac{a}{a+b}$$

2.4 Doob-Meyer theorem

Remark 19. This next theorem[4] provides a caraterization of submartingales.

Theorem 2.4.1 (Doob-Meyer theorem)**.** *(X_n) is a sub-martingale iff it can be written as sum of two sequences $(A_n), (M_n)$, that is $X_n = M_n + A_n, \forall n \geq 0$, such that:*

- *M_n is a martingale*
- *A_n is* increasing *(first two lines) and* predictable *(last) that is*

$$\begin{cases} A_0 = 0, \mathbb{E}[|A_n|] < +\infty, \forall n \\ A_0 \leq A_1 \leq A_2 \leq \dots \\ A_n \textit{ is } \mathscr{F}_{n-1}\textit{-measurable } \forall n \end{cases}$$

Moreover the sequences (M_n) and (A_n) are also almost surely unique.

Definition 2.4.1 (Predictability of sequence)**.** Predictability means that at time n our information $\mathscr{F}_n$ includes the value of A_{n+1} at the next time.

Dimostrazione. To prove $\Rightarrow$ let's suppose that (X_n) is a submartingale and define a sequence A_n such that $A_0 = 0$ and

$$A_n = \sum_{j=0}^{n-1} \left(\mathbb{E}[X_{j+1}|\mathscr{F}_j] - X_j\right)$$

We want to prove that A_n is predictable and increasing and that defining $M_n = X_n - A_n$ (so that $X_n = M_n + A_n$), M_n has to be a martingale. Let's start by showing A_n satisfies the conditions of the theorem:

[4]Doob was american; Meyer french and thus is pronounced french-like with accent on the second e (not german-like on first one).

- (A_n) is predictable: in the definition of A_n the index j arrives at $n-1$ so it involves the variable up to X_{n-1} and the first term up to $\mathbb{E}[X_n|\mathscr{F}_{n-1}]$. Hence all the involved elements are measurable with respect to $\mathscr{F}_{n-1}$ thus A_n is measurable with respect to $\mathscr{F}_{n-1}$ (or equivalently sequence (A_n) is predictable).

- (A_n) is increasing: given that (X_n) is a submartingale, by definition:

$$\begin{aligned}\mathbb{E}[X_{j+1}|\mathscr{F}_j] &\geq X_j, \quad \forall j\\ \mathbb{E}[X_{j+1}|\mathscr{F}_j] - X_j &\geq 0, \quad \forall j\end{aligned}$$

  And therefore A_n is increasing being

$$\begin{aligned}A_{n+1} &= \sum_{j=0}^{n} (\mathbb{E}[X_{j+1}|\mathscr{F}_j] - X_j)\\ &= \underbrace{\left(\sum_{j=0}^{n-1} (\mathbb{E}[X_{j+1}|\mathscr{F}_j] - X_j)\right)}_{A_n} + \underbrace{\mathbb{E}[(X_{n+1}|\mathscr{F}_n)] - X_n}_{\geq 0}\end{aligned}$$

  and so $A_{n+1} \geq A_n$.

Next define $M_n = X_n - A_n$ so obviously $X_n = M_n + A_n$. To show that (M_n) is a martingale:

$$\begin{aligned}\mathbb{E}[M_{n+1}|\mathscr{F}_n] &= \mathbb{E}[X_{n+1} - A_{n+1}|\mathscr{F}_n]\\ &= \mathbb{E}[X_{n+1}|\mathscr{F}_n] - \mathbb{E}[A_{n+1}|\mathscr{F}_n]\\ &\overset{(1)}{=} \mathbb{E}[X_{n+1}|\mathscr{F}_n] - A_{n+1}\\ &= \mathbb{E}[X_{n+1}|\mathscr{F}_n] - [A_n + (\mathbb{E}[X_{n+1}|\mathscr{F}_n] - X_n)]\\ &= X_n - A_n\\ &= M_n\end{aligned}$$

where in (1) the $\mathscr{F}_n$-measurability of A_{n+1} was used.
To prove $\Leftarrow$

$$\begin{aligned}\mathbb{E}[X_{n+1}|\mathscr{F}_n] &= \mathbb{E}[M_{n+1} + A_{n+1}|\mathscr{F}_n] = \mathbb{E}[M_{n+1}|\mathscr{F}_n] + \mathbb{E}[A_{n+1}|\mathscr{F}_n]\\ &\overset{(1)}{=} M_n + A_{n+1} \geq M_n + A_n = X_n\end{aligned}$$

where in (1), $\mathbb{E}[M_{n+1}|\mathscr{F}_n] = M_n$ being (M_n) a martingale and $\mathbb{E}[A_{n+1}|\mathscr{F}_n] = A_n$ being A_{n+1} $\mathscr{F}_n$ measurable.
On uniqueness we omit the demonstration. □

Remark 20. The following results show conditions under which a function $f : \mathbb{R} \to \mathbb{R}$ applied to a sequence (X_n) create a sequence $(f(X_n))$ which is submartingale

Theorem 2.4.2. *Let $f : \mathbb{R} \to \mathbb{R}$ be a real valued function such that $\mathbb{E}[|f(X_n)|] \leq \infty$ (the variable obtained applying the function has the mean). Now:*

1. *if (X_n) is a martingale and f is convex, OR ...*

2. *if (X_n) is a submartingale and f is convex and increasing ...*

... then the new sequence $(f(X_n))$ is a submartingale

Example 2.4.1. If (X_n) is a martingale, then $(|X_n|)$ is a submartingale (the absolute value is a convex function)

Example 2.4.2. If (X_n) is a martingale and the rv have the second moments $\mathbb{E}\left[X_n^2\right] < \infty, \forall n$, the sequence (X_n^2)

- is a submartingale (by point 1 of thm 2.4.2) since $f(x) = x^2$ is a convex function [5];
- thus by Doob-Meyer thm, since (X_n^2) is a submartingale we can write it as $X_n^2 = M_n + A_n$, with M_n a martingale and A_n increasing and predictable.

Example 2.4.3 (Special case)**.** Considering a random walk starting from 0 and defined as $X_n = 0 + \sum_{i=1}^n Z_i$ with Z_i iid and $\mathbb{E}[Z_1] = 0, \mathbb{E}\left[Z_1^2\right] = 1$.
Is it a martingale being a random walk with zero means Z_i (thm 2.2.1). Now let's define:

- $Y_n = X_n^2 - n$
- $\mathscr{F}_n = \sigma(Z_1, \ldots, Z_n)$

We prove that (Y_n) is still a martingale

$$\begin{aligned}
\mathbb{E}[Y_{n+1}|\mathscr{F}_n] &= \mathbb{E}\left[X_{n+1}^2 - (n+1)|\mathscr{F}_n\right] \\
&\overset{(1)}{=} \mathbb{E}\left[X_{n+1}^2|\mathscr{F}_n\right] - (n+1) \\
&= \mathbb{E}\left[(X_n + Z_{n+1})^2|\mathscr{F}_n\right] - (n+1) \\
&= \mathbb{E}\left[X_n^2 + Z_{n+1}^2 + 2X_n Z_{n+1}|\mathscr{F}_n\right] - (n+1) \\
&\overset{(2)}{=} X_n^2 + \mathbb{E}\left[Z_{n+1}^2|\mathscr{F}_n\right] + 2X_n\,\mathbb{E}[Z_{n+1}|\mathscr{F}_n] - (n+1) \\
&\overset{(3)}{=} X_n^2 + \mathbb{E}\left[Z_{n+1}^2\right] + 2X_n\,\mathbb{E}[Z_{n+1}] - (n+1) \\
&\overset{(4)}{=} X_n^2 - n \\
&= Y_n
\end{aligned}$$

where in:

- (1) we take the constant $(n+1)$ out of expectation
- (2) used the $\mathscr{F}_n$-measurability of X_n^2
- (3) by independence of Z_{n+1} from $\mathscr{F}_n$ we drop conditioning
- (4), we have that $\mathbb{E}\left[Z_{n+1}^2\right]$ is the variance and is equal to 1 and $\mathbb{E}[Z_{n+1}] = 0$ by assumption

Therefore being (Y_n) a martingale and since $X_n^2 = Y_n + n$, by the uniqueness in Doob-meyer thm one obtains $(M_n) = (Y_n)$ and $(A_n) = (n)$; this latter sequence (n) is both increasing (trivially: $1, 2, 3, \ldots$ is clearly increasing) and predictable (since is a constant).

[5] If (X_n) is a martingale (X_n^2) is a sub-martingale, but generally isn't even a martingale.

2.5 Martingale convergence

Remark 21. An important thm on martingales (and submartingales as well) says that if X_n is a martingale and satisfies some defined conditions it converge to a limit as $n \to \infty$. Before a refresher on convergences

Important remark 14 (RV convergence (refresher)). Let (X_n) be any sequence of real random variable on the probability space $(\Omega, \mathscr{A}, \mathbb{P})$. Almost sure and L_p convergences means

$$X_n \xrightarrow{a.s.} X \iff \mathbb{P}\left(\omega \in \Omega : \lim_n X_n(\omega) = X(\omega)\right) = 1$$

$$X_n \xrightarrow{L_p} X \iff \begin{cases} \mathbb{E}\left[|X_n|^p\right] < +\infty, \forall n \\ \mathbb{E}\left[|X|^p\right] < +\infty \\ \mathbb{E}\left[|X_n - X|^p\right] \to 0 \end{cases} \qquad \text{with } p > 0$$

Note that in particular for $p = 1$ and $p = 2$

$$X_n \xrightarrow{L_1} X \implies \mathbb{E}\left[|X_n|\right] \to \mathbb{E}\left[|X|\right]$$
$$X_n \xrightarrow{L_2} X \implies \mathbb{E}\left[X_n^2\right] \to \mathbb{E}\left[X^2\right]$$

Finally there's no connection between the two convergenges; they don't imply each other (different modes of convergence)

Theorem 2.5.1 (Martingale convergence thm)**.** *Let (X_n) be a submartingale:*

1. *if* $\sup_n \mathbb{E}\left[|X_n|\right] < \infty$, *then* $X_n \xrightarrow{a.s.} X$
2. *if* (X_n) *is* uniformly integrable, *then* $X_n \xrightarrow{a.s., L_1} X$
3. *if* (X_n) *is a martingale too and* $\sup_n \mathbb{E}\left[X_n^2\right] < \infty$ *then* $X_n \xrightarrow{a.s., L_1, L_2} X$

Remark 22. Some remarks:

- thm holds not only for martingale but also for submartingale (if a thm is valid for submartingale it's better);
- in the three points the conclusion is always the convergence of X_n, but the mode of convergence becomes better and better;
- in Point 1, the condition

  $$\sup_n \mathbb{E}\left[|X_n|\right] < +\infty \implies \mathbb{E}\left[|X_n|\right] < +\infty, \quad \forall n$$

  so all X_n have finite mean but the converse does not hold.
  In fact as counterexample suppose $\mathbb{E}\left[|X_n|\right] = n$, $\forall n$. Then each rv X_n has the mean but there's no constant that bounds $\mathbb{E}\left[|X_n|\right]$, so $\sup_n \mathbb{E}\left[|X_n|\right] = +\infty$. So in this case the first condition of theorem fails.
- Point 2 involvs *uniform integrability*: it's just a technical condition (don't attach an interpretation).

Definition 2.5.1 (Uniform integrability)**.** The sequence (X_n) is said uniformly integrable if and only if

$$\lim_{c \to +\infty} \sup_n \mathbb{E}\left[|X_n| \cdot \mathbb{1}(|X_n| > c)\right] = 0$$

Important remark 15. Remarks on uniform integrability

- we have that

  $$(X_n) \text{ uniformly integrable } \implies \sup_n \mathbb{E}\left[|X_n|\right] < +\infty$$

  but the converse is not true.
  Thus at point 2 of martingale convergence thm we are assume something more (uniform integrability) than what's needed for point 1; and thus obtain something more (not only a.s. convergence but also L_1).

- in can be shown that

  $$X_n \xrightarrow{L_1} X \iff \begin{cases} X_n \xrightarrow{p} X \\ (X_n) \text{ is uniformly integrable} \end{cases}$$

  Hence roughly speaking, uniformly integrability is what must be added to convergence in probability in order to reach L_1 convergence

Remark 23. Next we present a martingale (X_n) such that the $\sup_n \mathbb{E}\left[|(|X_n)\right] < \infty$ but (X_n) is not uniformly integrable. Thanks to martingale convergence we can say it converge a.s., but does not converge in L_1.

NB: Exam question: make an example of martingal satisfying the sup condition (point 1 of the convergence thm) but is not uniformly integrable. Then one has to provide one of the next two examples.

Example 2.5.1. Considering the symmetric random walk on the integers (starting from 1) and consequently it's a martingale (as the one starting from 0)

NB: Nella definizione di X_n, l'1 + l'ho messo io. . .

$$\begin{aligned} X_0 &= 1 \\ X_n &= 1 + \sum_{i=1}^{n} Z_i, \quad Z_i \text{ are iid} \\ \mathbb{P}(Z_1 = -1) &= \mathbb{P}(Z_1 = 1) = \frac{1}{2} \end{aligned}$$

Now considering the stopped sequence where (Y_n) copies (X_n) as long as this latter is positive and then when it become 0 repeats 0:

$$\begin{aligned} Y_n = X_{n \wedge \tau} &= \begin{cases} X_\tau & \text{if } n \geq \tau \\ X_n & \text{if } n < \tau \end{cases} \\ \tau &= \inf\{n \geq 0 : X_n = 0\} \end{aligned}$$

We now prove the three point regarding this new Y_n (not X_n):

1. (Y_n) is a martingale: if we stop a martingale we have a martingale again;

2. the sup condition 1 is verified: we have that $Y_n \geq 0$ and thus:

$$\mathbb{E}\left[|Y_n|\right] = \mathbb{E}\left[Y_n\right] \overset{(1)}{=} \mathbb{E}\left[Y_0\right] = \mathbb{E}\left[X_{\tau\wedge 0}\right] = \mathbb{E}\left[X_0\right] = \mathbb{E}\left[1\right] = 1$$

where (1) because expected value of martingale is constant.
Thus once again the sup condition is satisfied

NB: Be careful if X_n is a martin then $\mathbb{E}\left[X_n\right] = \mathbb{E}$ $\forall n$ but it's not true $\mathbb{E}\left[|X_n|\right] = \mathbb{E}\left[|X_0|\right]$.

$$\sup_n \mathbb{E}\left[|Y_n|\right] = 1 < +\infty$$

3. (Y_n) however is not uniformly integrable: being τ almost surely finite $(< +\infty)$:

$$Y_n = X_{n\wedge\tau} \xrightarrow{a.s.} X_\tau = 0$$

Hence if (Y_n) was uniformly integrable, we would obtain the following contradiction:

$$0 = \mathbb{E}\left[0\right] = \mathbb{E}\left[X_\tau\right] \overset{(1)}{=} \lim_n \mathbb{E}\left[Y_n\right] = \mathbb{E}\left[Y_0\right] = 1$$

in (1) by the martingale convergence thm and also it would be $Y_n \xrightarrow{a.s., L_1} X_\tau$

NB: salterei tenendo il precedente come esempio

NB: no qui l'1 non l'ho aggiunto ...

Example 2.5.2. Let

$$\begin{aligned}
&X_0 = 1\\
&X_n = e^{-\frac{n}{2}+\sum_{i=1}^n Z_i}\\
&Z_i \text{ iid} \sim \mathrm{N}\left(0,1\right)
\end{aligned}$$

We prove the sequence is a martingale, it satifies the sup condition but it's not uniformly integrable:

1. the sequence is a martingale: with the canonical filtration being $\mathscr{F}_n = \sigma(Z_1, \ldots, Z_n)$:

$$\begin{aligned}
\mathbb{E}\left[X_{n+1}|\mathscr{F}_n\right] &= \mathbb{E}\left[e^{-\frac{n+1}{2}} e^{\sum_{i=1}^{n+1} Z_i} | \mathscr{F}_n\right]\\
&= e^{-\frac{n+1}{2}} \cdot \mathbb{E}\left[e^{\sum_{i=1}^n Z_i} \cdot e^{Z_{n+1}} | \mathscr{F}_n\right]\\
&\overset{(1)}{=} e^{-\frac{n+1}{2}} \cdot e^{\sum_{i=1}^n Z_i} \cdot \mathbb{E}\left[e^{Z_{n+1}} | \mathscr{F}_n\right]\\
&\overset{(2)}{=} e^{-\frac{n+1}{2}+\sum_{i=1}^n Z_i} \cdot \mathbb{E}\left[e^{Z_{n+1}}\right]\\
&\overset{(3)}{=} e^{-\frac{n+1}{2}+\left(\sum_{i=1}^n Z_i\right)+\frac{1}{2}} = e^{-\frac{n}{2}+\sum_{i=1}^n Z_i}\\
&= X_n
\end{aligned}$$

where

- in (1) $e^{\sum_{i=1}^n Z_i}$ is measurable with respect to $\mathscr{F}_n$ so we take it out.
- in (2) because independence of the conditioning $\mathscr{F}_n$, we can drop the conditioning

- in (3) we note that $\mathbb{E}\left[e^{Z_{n+1}}\right]$ is the moment generating function of $\mathrm{N}(0,1)$ evaluated at point 1; so we look into a book and see that this is $e^{1/2}$

Thus the sequence is a martingale.

2. the $\sup \mathbb{E}[|X_n|] < \infty$ condition is verified: looking at X_n,

$$\mathbb{E}[|X_n|] \stackrel{(1)}{=} \mathbb{E}[X_n] \stackrel{(2)}{=} \mathbb{E}[X_0] = \mathbb{E}[1] = 1$$

where in:

- (1) being involved an exponential, it's always positive and so the absolute value of the expected value is superfluous;
- (2) because being a martingale expectation is constant

Hence:

$$\sup_n \mathbb{E}[|X_n|] = 1 \leq +\infty$$

3. finally the sequence is not uniformly integrable: proving by contradiction, suppose that (X_n) is univormy integrable; then by martingale convergence theorem one obtains $X_n \xrightarrow{a.s.} X$; but in this case it's easy to identify X, in fact:

NB: qui ci ho mollato, *eventualmente* rivedere

$$X_n = e^{-\frac{n}{2}+\sum_{i=1}^n Z_i} = e^{n\left(\frac{\sum_{i=1}^n Z_i}{n}-\frac{1}{2}\right)}$$

What is the limit of $\sum_{i=1}^n Z_i/n$? For the strong law of large number we have:

$$\frac{\sum_{i=1}^n Z_i}{n} \xrightarrow{a.s.} \mathbb{E}[Z_1] = 0$$

Hence:

$$\left(\frac{\sum_{i=1}^n Z_i}{n} - \frac{1}{2}\right) \xrightarrow{a.s.} -\frac{1}{2}$$
$$n\left(\frac{\sum_{i=1}^n Z_i}{n} - \frac{1}{2}\right) \xrightarrow{a.s.} -\infty$$
$$e^{n\left(\frac{\sum_{i=1}^n Z_i}{n}-\frac{1}{2}\right)} \xrightarrow{a.s.} e^{-\infty} = 0$$

So in this example the limit X of sequence X_n is actually 0 (dirac).
Now suppose (X_n) is uniformly integrable: then by point 2 of convergence thm

$$X_n \xrightarrow{a.s., L_1} X$$

and thus

$$0 = \mathbb{E}[0] = \mathbb{E}[X] \stackrel{(1)}{=} \lim_n \mathbb{E}[X_n] = \lim_n \mathbb{E}[1] = 1$$

where in (1) if uniformly integrable we have L_1 convergence so the equality is provided by L_1 convergence.
So this is a contradiction (absurd): assuming its uniformly integrable we obtain an absurdity. Then it's not uniformly integrable

Remark 24. Before another example let's see another consequence of the martingale convergence thm

Example 2.5.3. Let Z be a real random variable with mean $\mathbb{E}[|Z|] < +\infty$. Define

$$X_n = \mathbb{E}[Z|\mathscr{F}_n]$$

X_n is our prediction of Z at time n: X_n is a martingale infact

$$\mathbb{E}[X_{n+1}|\mathscr{F}_n] = \mathbb{E}[\mathbb{E}[Z|\mathscr{F}_{n+1}]|\mathscr{F}_n] \overset{(2)}{=} \mathbb{E}[Z|\mathscr{F}_n] \overset{(3)}{=} X_n$$

where in

- (2) just by applying the chain rule since $\mathscr{F}_n \subset \mathscr{F}_{n+1}$
- (3) by definition

So X_n is a martingale and it can be shown that X_n is uniformly integrable (take it as given); thus the convergence thm we can say that $X_n \xrightarrow{a.s.,L_1} X$ for some X. In this example we can also understand the form of X:

- X_n is our prediction of the information at time n.
- for $n \to +\infty$ it becomes $\mathbb{E}[Z|\mathscr{F}_\infty]$ that is the prediction of Z given the limit information (where $\mathscr{F}_\infty$ is the least σ-field which contains $\mathscr{F}_n$, $\forall n$)

Hence if Z is $\mathscr{F}_\infty$-measurable one obtains

$$X_n = \mathbb{E}[Z|\mathscr{F}_n] \xrightarrow{a.s.,L_1} \mathbb{E}[Z|\mathscr{F}_\infty] = Z$$

that is $X = Z$.
We discovered the hot water: in a sense this is an obvious fact. Indeed if Z is $\mathscr{F}_\infty$-measurable, then Z is known under the information $\mathscr{F}_\infty$. Hence it's reasonable that the prediction of Z under the information $\mathscr{F}_n$ converges to Z.

NB: bah

Example 2.5.4. Similarly to the last considerations, suppose (Y_n) is a sequence of real rv which converges almost surely to Y; let $\mathscr{F}_n = \sigma(Y_0, \dots, Y_n)$ be the sigma algebra/information generated by the first n variable and $Z = \lim_n Y_n$.
In this case $\mathbb{E}[Z|\mathscr{F}_n]$, our prediction of the limit given first n random variables: under the information $\mathscr{F}_\infty$ Z is known and the previous example shows that $\mathbb{E}[Z|\mathscr{F}_n] \xrightarrow{a.s.,L_1} Z$

Remark 25. As a final application of martingale convergence thm, let's prove the following result.

Proposition 2.5.2. *If (Z_n) is iid sequence with $\mathbb{E}[Z_1] = 0$, variance $\mathbb{E}\left[Z_1^2\right] = 1$ and τ is stopping time for the filtration $\mathscr{F}_n = \sigma(Z_1, \dots, Z_n)$ such that $\mathbb{E}[\tau] < +\infty$. Then mean and variance of sum of rvs, up to stopping time, are:*

$$\mathbb{E}\left[\sum_{i=1}^{\tau} Z_i\right] = 0$$

$$\text{Var}\left[\sum_{i=1}^{\tau} Z_i\right] = \mathbb{E}\left[\left(\sum_{i=1}^{\tau} Z_i\right)^2\right] = \mathbb{E}[\tau]$$

Remark 26. Before proving that what is the meaning? $\sum_{i=1}^{\tau} Z_i$ is the sum of a *random number* of *random variables*; suppose that $\tau = k$ (τ is constant dirac)

$$\mathbb{E}\left[\sum_{i=1}^{k} Z_i\right] = \sum_{i=1}^{k} \mathbb{E}\left[Z_i\right] = \sum_{i=1}^{k} 0 = 0$$
$$\text{Var}\left[\sum_{i=1}^{k} Z_i\right] = \sum_{i=1}^{k} \text{Var}\left[Z_i\right] = \sum_{i=1}^{k} 1 = k$$

So this thm generalizes the mean and variance of the sum of standardized indipendent random variables when τ is not a constant (trivial as seen above) but a random variable.

Dimostrazione. If we define $X_0 = 0$, $X_n = \sum_{i=1}^{n} Z_i$, (X_n) is a martingale (being a special random walk).
If we then define (Y_n) as

$$Y_0 = 0$$
$$Y_n = X_n^2 - n$$

we have that

- (Y_n) is a martingale as well (proved in example 2.4.3)
- $(Y_{n\wedge\tau})$ is still a martingale (it's a stopped martingale)

Out of the blue we could be interested in $\mathbb{E}\left[X_{n\wedge\tau}^2\right] - \mathbb{E}\left[n \wedge \tau\right]$, which is

$$\begin{aligned}\mathbb{E}\left[X_{n\wedge\tau}^2\right] - \mathbb{E}\left[n \wedge \tau\right] &= \mathbb{E}\left[X_{n\wedge\tau}^2 - (n \wedge \tau)\right] \\ &= \mathbb{E}\left[Y_{n\wedge\tau}\right] \overset{(1)}{=} \mathbb{E}\left[Y_{0\wedge\tau}\right] = \mathbb{E}\left[Y_0\right] \\ &= 0\end{aligned}$$

where (1) should be according to $(Y_{n\wedge\tau})$ being a martingale (with unmodified expected value). Thus looking at first and last term, it follows that

$$\mathbb{E}\left[X_{n\wedge\tau}^2\right] = \mathbb{E}\left[n \wedge \tau\right]$$

and thus by applying sup to both

$$\sup_n \mathbb{E}\left[X_{n\wedge\tau}^2\right] = \sup_n \mathbb{E}\left[n \wedge \tau\right] \overset{(1)}{=} \mathbb{E}\left[\tau\right] \overset{(2)}{<} +\infty$$

where (1) since the last is monotone and the sup of the last expectation is $\mathbb{E}\left[\tau\right]$ and (2) by assumption of thm.
Hence the martingale convergence thm implies that

$$X_{n\wedge\tau} \xrightarrow{a.s., L_1, L_2} X$$

On the other hand, since $\mathbb{E}\left[\tau\right] < +\infty$ then $\tau < +\infty$ almost surely which in turn implies $X_{n\wedge\tau} \xrightarrow{a.s.} X_\tau$ and thus $X = X_\tau$ almost surely.
To sum up

$$\mathbb{E}\left[\left(\sum_{i=1}^{\tau} Z_i\right)^2\right] = \mathbb{E}\left[X_\tau^2\right] = \lim_{n\to+\infty} \mathbb{E}\left[X_{n\wedge\tau}^2\right] = \lim_{n\to+\infty} \mathbb{E}\left[n \wedge \tau\right] = \mathbb{E}\left[\tau\right]$$

Moreover since we have L_1 convergence

$$\mathbb{E}\left[\sum_{i=1}^{\tau} Z_i\right] = \mathbb{E}\left[X_\tau\right] = \lim_{n\to+\infty} \mathbb{E}\left[X_{n\wedge\tau}\right] = \lim_{n\to+\infty} \mathbb{E}\left[X_{0\wedge\tau}\right] = \mathbb{E}\left[X_0\right] = 0$$

Finally

$$\begin{aligned}
\operatorname{Var}\left[\sum_{i=1}^{\tau} Z_i\right] &= \mathbb{E}\left[\left(\sum_{i=1}^{\tau} Z_i\right)^2\right] - \underbrace{\left[\mathbb{E}\left[\sum_{i=1}^{\tau} Z_i\right]\right]^2}_{0} \\
&= \mathbb{E}\left[\left(\sum_{i=1}^{\tau} Z_i\right)^2\right] \\
&= \mathbb{E}\left[\tau\right]
\end{aligned}$$

$\square$

Capitolo 3

Markov chains

3.1 Introduction

Definition 3.1.1 (Markov Chain (MC))**.** A sequence of real random variables (X_n) is a markov chain with respect to a filtration $(\mathscr{F}_n)$ if:

- (X_n) is adapted to $(\mathscr{F}_n)$
- $\mathbb{P}(X_{n+1} \in A|\mathscr{F}_n) = \mathbb{P}(X_{n+1} \in A|X_n)$, $\forall n \geq 0$, $\forall A \in \beta(\mathbb{R})$

Important remark 16*.* We have that:

- the second condition capture the idea of the markov chain: the probability of something regarding the future given all the information up to now depends only on actual value

  $$\mathbb{P}(\text{future}|\text{ present and the past}) = \mathbb{P}(\text{future}|\text{ present})$$

  Again, this is an assumption reasonable for some phenomenon, not all.

- it follows from the definition that going forward in the future more than one step depends still only on the value today:

  $$\mathbb{P}(X_{n+1} \in A_1, \ldots, X_{n+k} \in A_k|\mathscr{F}_n) = \mathbb{P}(X_{n+1} \in A_1, \ldots, X_{n+k} \in A_k|X_n)$$

- Often for n fixed, considered filtration is just $\mathscr{F}_n = \sigma(X_0, X_1, \ldots, X_n)$, that is the know value of all Xs up to X_n $(X_0 = x_0, X_1 = x_1, \ldots)$.

Definition 3.1.2 (Kernel of a Markov chain)**.** It's the probability of a future event given the actual value:

$$\alpha_n(x, A) = \mathbb{P}(X_{n+1} \in A|X_n = x), \quad \forall n \geq 0, \forall x \in \mathbb{R}, \forall A \in \beta(\mathbb{R})$$

Definition 3.1.3 (Homogeneous MC)**.** A Markov chain is said to be homogeneous if its kernel is constant: $\alpha_n = \alpha_0$, $\forall n \geq 0$.

Remark 27 (Remarks)*.* We have that

- informally the kernel $\alpha_n(x, A)$ is the probability of entering the set A starting from x at time n

- if the MC is *homogeneous* that probability is always the same ($\alpha_n = \alpha_0, \forall n \geq 0$) and for this reason eg

$$\mathbb{P}\left(X_{n+1} \in A | X_n = x\right) = \mathbb{P}\left(X_1 \in A | X_0 = x\right), \quad \forall n \geq 0$$

  being the α index superfluous we just write α to mean the kernel

- if the Markov chain is homogeneous, to write down the distribution of any X_n in the sequence we need only a starting point (random variable) and the kernel. Thus to fully describe the full sequence (X_n) just the distribution of X_0 and α;

- in the rest of this course we'll focus only on *homogeneous* Markov chains, which are the most important type. If we forget to say homogeneous, this is implied

Example 3.1.1 (Randow walk as homogeneous markov chain)**.** Any random walk is an homogeneous Markov chain (not always a martingale, but always a Markov chain).
Consider a random walk

$$X_n = X_0 + \sum_{i=1}^{n} Z_i, \quad \text{with } Z_i \text{ iid and } X_0 \perp\!\!\!\perp (Z_i)$$

To prove it's a Markov chain we calculate

$$\begin{aligned}
\mathbb{P}\left(X_{n+1} \in A | X_0 = x_0, \ldots, X_n = x_n\right) &= \mathbb{P}\left(X_n + Z_{n+1} \in A | X_0 = x_0, \ldots, X_n = x_n\right) \\
&\overset{(1)}{=} \mathbb{P}\left(x_n + Z_{n+1} \in A | X_0 = x_0, \ldots, X_n = x_n\right) \\
&\overset{(2)}{=} \mathbb{P}\left(x_n + Z_{n+1} \in A\right) \\
&\overset{(3)}{=} \mathbb{P}\left(x_n + Z_1 \in A\right)
\end{aligned}$$

where in:

- (1) we replaced with x_n at left of conditioning since known
- (2) we noted Z_{n+1} are independent from $X_0, \ldots, X_n$ so we drop conditioning
- (3) being identically distributed

Now suppose we want to evaluate the probability conditioning only on the last value; by repeating the same argument as above we have

$$\mathbb{P}\left(X_{n+1} \in A | X_n = x_n\right) = \mathbb{P}\left(x_n + Z_1 \in A\right)$$

So we proved that

$$\mathbb{P}\left(X_{n+1} \in A | X_0 = x_0, \ldots, X_n = x_n\right) \overset{(1)}{=} \mathbb{P}\left(X_{n+1} \in A | X_n = x_n\right) = \mathbb{P}\left(x_n + Z_1 \in A\right)$$

where the (1) equality means this is a markov chain.
Furthermore the kernel does not depend on time making it an homogeneous markov chain

$$\alpha(x, A) = \mathbb{P}\left(X_1 \in A | X_0 = x\right) = \mathbb{P}\left(x + Z_1 \in A\right)$$

3.2 Stationarity

Definition 3.2.1 (Stationarity of a generic sequence)**.** A sequence or rv (X_n) (not necessarily a Markov chain) is said stationary if the probability distribution doesn't change if we shift the sequence by 1

$$(X_1, X_2, \ldots) \sim (X_0, X_1, \ldots)$$

Important remark 17 (Stationarity and identically distributed rvs)*.* By induction[1] if follows that for a stationary sequence if we shift by k we still get the same distribution

$$(X_0, X_1, \ldots) \sim \ldots \sim (X_k, X_{k+1}, \ldots)$$

so that

$$X_0 \sim X_k, \quad \forall k$$

Hence

$$(X_n) \text{ is stationary} \implies X_i \text{ are identically distributed } \forall i$$

The converse does not generally hold. A special case where it does is markov chains.

Theorem 3.2.1. *If (X_n) is a MC then*

$$(X_n) \textit{ is stationary} \iff X_i \textit{ are identically distributed } \forall i \iff X_1 \sim X_0$$

Definition 3.2.2 (Stationary distribution for a Markov chain)**.** Let π be a probability measure on $\beta(\mathbb{R})$; it is called stationary distribution for a MC (X_n) if

$$X_0 \sim \pi \implies X_1 \sim \pi$$

Important remark 18*.* Thus if (X_n) has stationary distribution π for thm 3.2.1

$$X_0 \sim \pi \sim X_1 \implies (X_n) \text{ is stationary}$$

Example 3.2.1 (IID sequence as MC)**.** A trivial example of MC is an iid sequence of rvs. Infact, if (X_n) is iid, then:

$$\mathbb{P}\left(X_{n+1} \in A | X_0, \ldots, X_n\right) \stackrel{(1)}{=} \mathbb{P}\left(X_{n+1} \in A\right) \stackrel{(2)}{=} \mathbb{P}\left(X_0 \in A\right)$$

with (1) for independence and (2) for identical distribution.
In this case the kernel α is

$$\alpha(x.X) = \mathbb{P}\left(X_0 \in A\right), \quad \forall x \in \mathbb{R}, \forall A \in \beta(\mathbb{R})$$

[1] If shifting by one doesn't change the distribution then even if we start from the shifted sequence.

3.3 Discrete Markov chains

Important remark 19. From now on:

- we assume that X_n are discrete rvs $\forall n \geq 0$, taking value in a finite or countable set S;
- set S is called the *state space* of the Markov chain; its elements are called *states*;
- kernel notation simplifies. Instead of writing the general

  $$\alpha(x, A) = \mathbb{P}\left(X_1 \in A | X_0 = x\right)$$

  we can write

  $$\alpha(x, y) = \mathbb{P}\left(X_1 = y | X_0 = x\right) = \mathbb{P}\left(\text{going from } x \text{ to } y \text{ in one step}\right)$$

NB: non fatto quest'anno

Example 3.3.1 (Modelling political behaviour (party choosen at election))**.** If we suppose the vote at a given election follows an homogeneous Markov chain the probability to choose a given party Y depends only on the first time we voted plus the kernel

$$\mathbb{P}\left(\text{I select } Y | \text{I selected } X \text{ in the previous election}\right) = a$$
$$\mathbb{P}\left(X_0\right) = \begin{cases} \text{prob I choose } X \text{ at the first election I participated} \\ \text{prob I choose } Y \text{ at the first election I participated} \\ \text{prob I choose } Z \text{ at the first election I participated} \end{cases}$$

Here

$$S = \{\text{set of all party that can be selected}\} = \{x, y, z\}$$

The kernel is

$$\alpha(x, y) = \mathbb{P}\left(\text{I select } y \mid \text{at previous selected } x\right)$$

and the distribution at time X_0 being

$$\mathbb{P}\left(X_0 = x\right)$$
$$\mathbb{P}\left(X_0 = y\right)$$
$$\mathbb{P}\left(X_0 = z\right)$$

Example 3.3.2. Consider the symmetric random walk on $\mathbb{Z}$

$$X_0 \in \mathbb{Z}, \text{ almost surely}$$
$$X_n = X_0 + \sum_{i=1}^{n} Z_i, X_0 \perp\!\!\!\perp Z_i, Z_i \text{ is iid}$$
$$\mathbb{P}\left(Z_1 = -1\right) = \mathbb{P}\left(Z_1 = 1\right) = \frac{1}{2}$$

In this case $S = \mathbb{Z}$.
What is the kernel? Just write down the definition/formula to help with developing the kernel:

$$\alpha(x, y) = \mathbb{P}(X_1 = y | X_0 = x) = \begin{cases} \frac{1}{2} & \text{if } y = x + 1\text{, we increment by 1} \\ \frac{1}{2} & \text{if } y = x - 1\text{, we decrement by 1} \\ 0 & \text{otherwise} \end{cases}$$

Definition 3.3.1 (Stopping time for state x)**.** For each state $x \in S$ let

$$\begin{aligned} \tau_x &= \inf \{n > 0 : X_n = x\} \\ &= \begin{cases} +\infty & \text{if } X_n \neq x,\ \forall n \geq 0 \\ \text{first time } n \text{ such that } X_n = x & \text{otherwise} \end{cases} \end{aligned}$$

Definition 3.3.2 (Recurrent state)**.** State $x \in S$ is called recurrent if

$$\mathbb{P}(\tau_x < +\infty | X_0 = x) = 1$$

That is loosely speaking: if we start from it, sooner or laterif we are sure to return back to it.

Proposition 3.3.1. *It can be shown that if state x is* recurrent *then*

$$\mathbb{P}(X_n = x \text{ for infinitely many } n | X_0 = x) = 1$$

That is, if we start from a recurrent state not only we go back to it one time almost surely, but we'll return back to it again and again almost surely.
Idea: being in a Markov chain we forget about the past. We're sure that if we start from a point we will back to it; once this happens, you're in the state for the second time and working with a Markov chain you forget the past so you'll be back on the state a third time. An so on;

Proposition 3.3.2. *It can be shown that*

$$x \text{ is a recurrent state} \iff \sum_n \mathbb{P}(X_n = x | X_0 = x) = +\infty$$

Definition 3.3.3 (Transient state)**.** State $x \in S$ is called transient if not recurrent

$$\mathbb{P}(\tau_x < +\infty | X_0 = x) < 1$$

Remark 28. If the state is recurrent, i'm sure that sooner or later i'll be back to it, but the time this happens can be quite different and so we two cases

Definition 3.3.4 (Null and positively recurrent)**.** If x is recurrent:

1. x is said null recurrent if

   $$\mathbb{E}[\tau_x | X_0 = x] = +\infty$$

   that is the expected time we come back is long

2. x is said positively recurrent if

$$\mathbb{E}\left[\tau_x | X_0 = x\right] < +\infty$$

that is the expected/mean time when we come back is (at least) finite

Example 3.3.3 (Funny example)**.** Both happy days and bad days are recurrent states (they come back sooner or later) but unfortunately bad days are positively recurrent while happy days are null recurrent.

Important remark 20. Some other useful facts about states types (which can be shown):

1. if state y is *transient* or *null recurrent*, then

$$\lim_{n \to +\infty} \mathbb{P}\left(X_n = y | X_0 = x\right) = 0, \forall x$$

whichever starting point x probability to come back to it tends to go to 0 (it's not 0);

2. if S *is finite* (remember S can be finite or countable) then

 - there are no null recurrent state ($\nexists$)
 - exists at least one positive recurrent state.

Eg in the case of symmetric rw on $\mathbb{Z}$ all states are null recurrent.

Definition 3.3.5 (Irreducible Markov chain)**.** A Markov chain is irreducible if for any couple of states $\forall x, y \in S$, $\exists m, n$ such that

$$\mathbb{P}\left(X_m = y | X_0 = x\right) > 0 \text{ and } \mathbb{P}\left(X_n = x | X_0 = y\right) > 0$$

Remark 29 (Interpretation). If irreducible, every state must be reachable from every other state

Proposition 3.3.3. *If a chain is irreducible all states are of the same type: all are either transient, null recurrent or positive recurrent.*

Example 3.3.4. If $S = \{0, 1, 2, \ldots\}$ and $\alpha(n, n+1) = 1$ then the MC is not irreducible (we can go only from n to $n+1$) but all states are transient.

Example 3.3.5. If $S = \{a, b\}$, and $\alpha(a, b) = 1 = \alpha(b, a)$ then the MC is irreducible and positive recurrent.

Example 3.3.6 (Symmetric random walk on on **Z** irreducibility and states)**.** Symmetric random walk ($X_n = X_0 + \sum_{i=1}^{n} Z_i$, $X_0 \perp\!\!\!\perp Z_i$, Z_i and $\mathbb{P}\left(Z_i = -1\right) = \mathbb{P}\left(Z_i = 1\right) = \frac{1}{2}$) is irreducibile and null recurrent:

- *irreducibility*: supposing at a certain time we are in a point x and we want to arrive to a point y. Supposing $x < y$, if we want to go from x to y it's enough to observe 1 $(y - x)$ times (and this has a strictly positive probability); otherwis if we want to go from y to x it's enough to observe -1 exactly $(y - x)$ times (which has a positive probability). So the states all either transient, null recurrent or positive recurrent

- *recurrency*: since all states are of one type let's consider the state $x = 0$ and try to determine its type. We apply proposition 3.3.2 to determine it's recurrent, so we start evaluating

$$\begin{aligned}
\sum_n \mathbb{P}\left(X_n = 0 | X_0 = 0\right) &\overset{(1)}{=} \sum_{n \text{ even}} \mathbb{P}\left(X_n = 0 | X_0 = 0\right) \\
&= \sum_n \mathbb{P}\left(X_{2n} = 0 | X_0 = 0\right) \\
&\overset{(2)}{=} \sum_n \mathbb{P}\left(X_0 + \sum_{i=1}^{2n} Z_i = 0 | X_0 = 0\right) \\
&= \sum_n \mathbb{P}\left(\sum_{i=1}^{2n} Z_i = 0\right) \\
&= \sum_n \mathbb{P}\left(\sum_{i=1}^{2n} \frac{Z_i + 1}{2} = n\right) \\
&\overset{(3)}{=} \sum_n \mathbb{P}\left(\text{Bin}\left(2n, \frac{1}{n}\right) = n\right) \\
&= \sum_n \left[\binom{2n}{n}\left(\frac{1}{2}\right)^{2n}\right] \\
&\overset{(4)}{=} \sum_n \frac{C_n}{\sqrt{n}} \\
&= +\infty
\end{aligned}$$

where in

  - (1) we can restrict indexes of the sum because if n is odd $\mathbb{P}\left(X_n = 0 | X_0 = 0\right) = 0$ (at each step we can go up or low by 1, if the steps are odd we will never come back to 0);
  - (2) by substituting the random walk definition
  - (3) since $\frac{Z_i+1}{2} \in \{0, 1\}$ are iid with $\sum_{i=1}^{2n} \frac{Z_i+1}{2}$ we're basically summing several independent bernoulli rvs so the sum is a binomial and especially $\sum_{i=1}^{2n} \frac{Z_i+1}{2} \sim \text{Bin}\left(2n, \frac{1}{2}\right)$;
  - in (4) we go to the friendly mathematician which uses Sterling formula ad this sums up to $+\infty$ so $\sum_n \mathbb{P}\left(X_n = 0 | X_0 = x\right) = +\infty$.

  So we proved that 0 is a recurrent state and therefore all the states (being the same type being the chain irreducible) are recurrent

- now we prove that 0 is null recurrent by contradiction.
  Consider $\mathbb{E}\left[\tau_0 | X_0 = 0\right]$ with $\tau_0 = \inf\{n > 0 : X_n = 0\}$ (so it's the first positive time we are in 0), we have two possible situations:

$$\mathbb{E}\left[\tau_0 | X_0 = 0\right] = \begin{cases} < \infty & \text{if true, 0 is positive recurrent} \\ = +\infty & \text{if true, 0 is null recurrent} \end{cases}$$

  Toward a contradiction suppose that the expected time is finite ($\mathbb{E}\left[\tau_0 | X_0 = 0\right] < \infty$), so 0 is positive recurrent.

We now apply proposition 2.5.2, (application of the martingale convergence theorem) stating that if (Z_n) is iid sequence with $\mathbb{E}[Z_1] = 0$, variance $\mathbb{E}[Z_1^2] = 1$ and τ is stopping time for the filtration $\mathscr{F}_n = \sigma(Z_1, \dots, Z_n)$ such that $\mathbb{E}[\tau] < +\infty$ mean and variance of sum of rvs, up to stopping time, are:

$$\mathbb{E}\left[\sum_{i=1}^{\tau} Z_i\right] = 0$$
$$\operatorname{Var}\left[\sum_{i=1}^{\tau} Z_i\right] = \mathbb{E}\left[\left(\sum_{i=1}^{\tau} Z_i\right)^2\right] = \mathbb{E}[\tau]$$

Using this result in this case sum and variance the expected value of the process at stopping time are

$$\begin{aligned}
\mathbb{E}[X_{\tau_0} | X_0 = 0] &= 0 \\
\operatorname{Var}[X_{\tau_0} | X_0 = 0] &= \mathbb{E}\left[X_{\tau_0}^2 | X_0 = 0\right] \\
&= \mathbb{E}\left[\left(0 + \sum_{i=1}^{\tau_0} Z_i\right)^2 | X_0 = 0\right] \\
&\overset{(1)}{=} \mathbb{E}[\tau_0 | X_0 = 0]
\end{aligned}$$

where (1) since $\mathbb{E}[\tau_0 | X_0 = 0] < +\infty$.
Since the stopping time $\tau_0 \geq 1$ then it expected value

$$\mathbb{E}[\tau_0 | X_0 = 0] \geq 1$$

At the same time the last expression is the variance of the value when it comes back to 0 so it would seem that

$$\operatorname{Var}[X_{\tau_0} | X_0 = 0] = \mathbb{E}\left[X_{\tau_0}^2 | X_0 = 0\right] \geq 1$$

This has to be a contradition because $X_{\tau_0} = 0$ almost surely (and its variance cannot be positive): since $\tau_0 = \inf\{n > 0 : X_n = 0\}$ is the first time the process come back to zero, then $X_{\tau_0} = 0$ by definition (it's a actually degenerate random variable with null variance and variance can be greater than 1).
This proves that 0 is null recurrent and also all the states are.

3.4 Generalized symmetric random walk on $\mathbb{Z}^k$

Remark 30. Let's briefly generalize the symmetric random walk on $\mathbb{Z}^k$.

Definition 3.4.1 (Symmetric random walk on $\mathbb{Z}^k$)**.** It is defined as

$$\mathbf{X}_n = \mathbf{X}_0 + \sum_{i=1}^{n} \mathbf{Z}_i$$

with

- $\mathbf{X}_0, \mathbf{Z}_i \in \mathbb{Z}^k$ (so we're summing random vectors)
- $\mathbf{X}_0 \perp\!\!\!\perp \mathbf{Z}_i$
- $(\mathbf{Z}_i)$ iid where the single increment vector is

$$\mathbb{P}(\mathbf{Z}_1 = \mathbf{x}) = \frac{1}{2k}, \quad \forall \mathbf{x} \in \bigcup_{i=1}^{k} \left\{ \mathbf{y} \in \mathbb{Z}^k : y_i \in \{-1, 1\}, y_j = 0 \forall j \neq i \right\}$$

  The set on the right is composed by vectors $\mathbf{x}$ where one components (in turn) is either -1 or 1 and all the remainings are 0.

Example 3.4.1. For instance:

- if $k = 1$, $\mathbb{P}(Z_1 = 1) = \mathbb{P}(Z_1 = -1) = \frac{1}{2}$ (that is in $\mathbb{Z}^1$ there are only two deltas available, $\{-1, +1\}$ each with probability $1/2$, as seen before)
- if $k = 2$, $\mathbb{P}(\mathbf{Z}_1 = \mathbf{x}) = \frac{1}{4}$: in $\mathbb{Z}_2$ there are four admisible values/points/deltas are $\{(0, -1), (0, 1), (-1, 0), (1, 0)\}$ each with the same probability $1/4$. In this case we can move either north, south, east or west in the cartesian plane at each step.

Remark 31 (Type of random walk). This random walk is used often in application and gives origin to a strange phenomenon. The symmetric random walk on $\mathbb{Z}^k$ is

- still **irreducible** (can be proved by exactly the same argument we done with $k = 1$), so all states are of the same type.
- regarding type of events it can be shown that [2]:
  - for $k = 1$ or $k = 2$ it is *null recurrent*;
  - for $k \geq 3$ it's *transient*.

NB: Exam question: can you give me an exameple of chain that is irreducible and transient? This is the only example in the course: it is the symmetric random walk on $\mathbb{Z}^k$ with $k \geq 3$.

3.5 Stationary distributions

Remark 32. A probability π on S is said to be *stationary distribution* if

$$X_0 \sim \pi \implies X_1 \sim \pi$$

Theorem 3.5.1 (Stationary Markov chain)**.** *Let (X_n) be a Markov chain; if $X_0 \sim \pi$ and π is a stationary distribution ($X_0 \sim \pi \implies X_1 \sim \pi$), then the sequence $(X_0, X_1, \ldots)$ is stationary (in the sense that $(X_1, X_2, \ldots) \sim (X_5, X_6, \ldots)$)*

Important remark 21. Let's discuss the existence of a stationary distribution. Stationary distribution are interesting objects: but given a Markov chain we're not sure whether a stationary distribution exists. And thus is useful a thm which explain in which situations a stationary distribution exists.

[2]This is a curios fact: the type change drastically with the number of dimensions. If $k = 2$ and we start from a point eg, if $\mathbf{X}_0 = (i, j)^\mathsf{T}$ then with probability 1 we will go back to this point; instead if $k \geq 3$ we start from a point but it could be that we never go back there (we don't prove it).

Theorem 3.5.2. *If $X_n \in S, \forall n \geq 0$, where S is finite or coutable then*

$$\exists \text{ stationary distribution} \iff \exists \text{ a positive recurrent state}$$

Moreover if the Markov chain is irreducible and a stationary distribution exists, then it is unique.

Proposition 3.5.3. *It can be shown that, if S is finite, there's at least one positive recurrent state (and there are no null recurrent state); thus there is certainly a stationary distribution*

Remark 33. Before moving to examples it is useful to have a criterion to tell if a distribution is stationary.

Theorem 3.5.4. *A probability π on S is a stationary distribution if and only if*

$$\pi(b) = \sum_{a \in S} \pi(a)\alpha(a,b), \quad \forall b \in S$$

TODO: a me sembra abbia dimostrato solo un lato dell'implicazione

Dimostrazione. we have:

- If π is a stationary distribution then, by definition

  $$\mathbb{P}(X_0 = b) = \mathbb{P}(X_1 = b) = \pi(b), \quad \forall b \in S$$

  at the same time

  $$\begin{aligned}\pi(b) = \mathbb{P}(X_1 = b) &= \sum_{a \in S} \mathbb{P}(X_0 = a) \cdot \mathbb{P}(X_1 = b | X_0 = a) \\ &= \sum_{a \in S} \pi(a) \cdot \alpha(a,b)\end{aligned}$$

- ?

□

Remark 34. In practice to check whether a distribution π is stationary we have to check the previous equation.

NB: non fatto quest'anno

Example 3.5.1 (Trivial but useful example)**.** Suppose S consist of two point

$$S = \{a, b\}, \quad \alpha(a,b) = \alpha(b,a) = 1$$

In this Markov chain we go from one point to the other with probability 1, and then we go back with probability 1.
Is the chain irreducible? yes. Furthermore S is finite so there's one and only one stationary distribution.
Who is the unique stationary distribution? In this example the unique stationary distribution could be $\pi(a) = \pi(b) = 1/2$ (it's a conjecture, we don't expect the two states have different probability). To verify it:

$$\sum_{x \in S} \pi(x) \cdot \alpha(x, a) = \underbrace{\pi(a)}_{1/2}\underbrace{\alpha(a,a)}_{0} + \underbrace{\pi(b)}_{1/2}\underbrace{\alpha(b,a)}_{1} = \frac{1}{2} = \pi(a)$$

$$\sum_{x \in S} \pi(x) \cdot \alpha(x, b) = \underbrace{\pi(a)}_{\frac{1}{2}}\underbrace{\alpha(a,b)}_{1} + \underbrace{\pi(b)}_{\frac{1}{2}}\underbrace{\alpha(b,b)}_{0} = \frac{1}{2} = \pi(b)$$

Example 3.5.2 (Ehrenfest urn)**.** We have two boxes and r balls, some in the first remaining in the second. At each time $n \geq 0$ we select a ball at random (each with probability $1/r$) and moves from the urn where it is to the other [3]. Let X_n = "number of balls in box 1 at time n"; the number of balls can be $0, 1, \ldots$, so $S = \{0, 1, \ldots, r\}$. The kernel is

$$\begin{aligned}
\alpha(a, b) &= \mathbb{P}\left(X_1 = b | X_0 = a\right) \\
&= \begin{cases} \frac{a}{r} & \text{if } b = a - 1 \text{ (we draw from this box which decreases)} \\ 1 - \frac{a}{r} & \text{if } b = a + 1 \text{ (we draw from the other box, so increases)} \\ 0 & \text{otherwise} \end{cases}
\end{aligned}$$

In this case a stationary distribution π exists (since S is finite) and it's unique (since the chain is irreducible). This time it's not obvious, but the stationary distribution is defined as:

$$\pi(b) = \frac{\binom{r}{b}}{2^r}, \quad \forall b \in S = \{0, 1, \ldots, r\}$$

As exercise we now verify that it is actually the stationary distribution. We have to check

$$\pi(b) = \sum_{a \in S} \pi(a) \cdot \alpha(a, b), \quad \forall b \in S$$

In case of ehrenfest urn we have:

$$\begin{aligned}
&\pi(b) = \frac{\binom{r}{b}}{2^r} \\
&\alpha(a, b) = 0 \quad \text{if } a \notin \{b - 1, b + 1\}
\end{aligned}$$

and thus we get

$$\begin{aligned}
\sum_{a \in S} \pi(a)\alpha(a, b) &= \pi(b - 1) \cdot \alpha(b - 1, b) + \pi(b + 1) \cdot \alpha(b + 1, b) \\
&= \frac{\binom{r}{b-1}}{2^r} \cdot \left(1 - \frac{b - 1}{r}\right) + \frac{\binom{r}{b+1}}{2^r} \cdot \frac{b + 1}{r} \\
&= \frac{1}{2^r}\left[\frac{r!}{(b - 1)!(r - b + 1)!} \cdot \frac{r - b + 1}{r} + \frac{r!}{(b + 1)!(r - b - 1)!} \cdot \frac{b + 1}{r}\right] \\
&= \frac{1}{2^r}\left[\frac{(r - 1)!}{(b - 1)!(r - b)!} + \frac{(r - 1)!}{(b)!(r - b - 1)!}\right] \\
&= \frac{1}{2^r}\left[\frac{(r - 1)!}{(b - 1)!(r - b)(r - b - 1)!} + \frac{(r - 1)!}{(b)(b - 1)!(r - b - 1)!}\right] \\
&= \frac{1}{2^r}\frac{(r - 1)!}{(b - 1)!(r - b - 1)!}\underbrace{\left[\frac{1}{r - b} + \frac{1}{b}\right]}_{\frac{r}{b(r-b)}} \\
&= \frac{1}{2^r}\frac{r!}{b!(r - b)!} \\
&= \frac{\binom{r}{b}}{2^r} \\
&= \pi(b)
\end{aligned}$$

[3] This simple model is useful to describe termodynamical stuff/systems

Incidentally note that π is a probability measure on S since

$$\sum_{b\in S}\pi(b)=\frac{\sum_{b\in S}\binom{r}{b}}{2^r}=\frac{2^r}{2^r}=1$$

NB: bah

Example 3.5.3. For symmetric random walk on $\mathbb{Z}$ a stationary distribution does not exist since all states are null recurrent.
However letting

$$\pi(b)=1,\quad \forall b\in\mathbb{Z}$$

it can be shown that

$$\pi(b)=\sum_{a\in\mathbb{Z}}\pi(a)\alpha(a,b),\quad \forall b\in\mathbb{Z}$$

This is not a contradiction since π is NOT a probability measure on $\mathbb{Z}$. Indeed

$$\sum_{b\in\mathbb{Z}}\pi(b)=\sum_{b\in\mathbb{Z}}1=+\infty$$

3.6 Ergodicity

Definition 3.6.1 (Ergodic Markov chain)**.** A Markov chain is said to be ergodic if exists a probability measure π on S such that:

$$\pi(b)=\lim_{n\to+\infty}\mathbb{P}\left(X_n=b|X_0=a\right),\quad \forall a,b\in S$$

Important remark 22. Two remarks:

1. interpretation[4]: if the MC is ergodic in a sense the MC forgets the starting state, that is probability of arriving to the state b (somewhere) in the limit does not depend on the state a from which we started (starting point).
2. if the Markov chain is ergodic, it can be shown that π is the *only stationary distribution*;

Remark 35. Now we'll see some results: first a criterion that implies that the chain is actually ergodic.

Theorem 3.6.1 (Sufficient condition for ergodicity)**.** *If a chain is irreducible, aperiodic and positive recurrent, then it is ergodic.*

Definition 3.6.2 (Aperiodic state)**.** A state x is said to be aperiodic if the set of n such that

$$\{n:\mathbb{P}\left(X_n=x|X_0=x\right)=0\}$$

is finite. Then if x is aperiodic one obtains

$$\mathbb{P}\left(X_n=x|X_0=x\right)>0,\quad \forall \text{ sufficiently large } n$$

[4] Actually "ergodicity" has different meanings in different framework. In fact, there are problems in probability theory where "ergodicity" has a different meaning with respect to the above"

Important remark 23. If the MC is irreducible all states are either aperiodic or not aperiodic: that is all the states are of the same type even considered this kind of classification.

Example 3.6.1. The symmetrical random walk on $\mathbb{Z}$ is not aperiodic. Infact $\forall a \in S$

$$\{n : \mathbb{P}\left(X_n = a | X = a\right) = 0\} = \{\text{odd integers}\}$$

(eg if we start from 0 we can go back to 0 at odds times) and the set of odd integers is not finite. Having proved for an example 0 and knowing chain being irreducible, all states are not aperiodic.

Example 3.6.2. if $S = \{a, b\}$ and $\alpha(a, b) = \alpha(b, a) = 1$ then similarly $\{n : \mathbb{P}\left(X_n = a | X = a\right) = 0\} = \{\text{odd integers}\}$ and it's not aperiodic.

Example 3.6.3. The symmetric RW on integers is *not ergodic*: the definition on ergodicity the probability π we should obtain in the lim is the unique stationary distribution, but the symmetric random walk on the integers does not have any stationary distribution because all states are null recurrent. And the existence of a stationary distribution is equivalent to the existence of a positive recurrent state. So being all states null recurrent there are no positive recurrent. Thus the symmetric rw does not admit any stationary distribution, and in particular is not ergodic, because if it was ergodic the probability π would be the unique distribution.

NB: detto gli anni scorsi, se mai riscrivi/riordina

3.7 Exchangeability

3.7.1 Stationary and exchangeable sequences

Remark 36. Stationary sequences and exchangeable sequences are important sequences of random variables:

- Stationary is important/common assumption for time time series [5]
- Exchangeable sequences were introduced by Bruno de Finetti; it's a nice assumption for a sequence, applied in many fields, especially bayesian statistics.

Definition 3.7.1 (Stationary sequences (ripasso))**.** Let $(X_0, X_1, X_2, \ldots)$ be a sequence of real random variables. The sequence (X_n) is stationary if and only if the distribution doesn't change if we shift it by one, eg $(X_1, X_2, X_3, \ldots) \sim (X_0, X_1, X_2, \ldots)$. By induction one obtains that

$$(X_k X_{k+1}, X_{k+2}, \ldots) \sim (X_0, X_1, X_2, \ldots)$$

so we can shift the sequence by $k \geq 1$.

Definition 3.7.2 (Exchangeable sequences)**.** A sequence (X_n) is exchangeble by definition iff $(X_{\pi_0}, X_{\pi_1}, \ldots, X_{\pi_n}) \sim (X_0, X_1, \ldots, X_n)$ $\forall n \geq 1$, $\forall$ permutation $(\pi_0, \pi_1, \ldots, \pi_n)$ of $(0, 1, \ldots, n)$.

[5] statistican call time series sequences of random variables

Example 3.7.1. So:

- if $n = 1$ it must be that $(X_0, X_1) \sim (X_1, X_0)$
- if $n = 2$ the permutation are $3! = 6$ and must be:

$$(X_0, X_1, X_2) \sim (X_0, X_2, X_1) \sim (X_2, X_0, X_1) \sim (X_1, X_0, X_2) \sim (X_1, X_2, X_0) \sim (X_2, X_1, X_0)$$

3.7.2 Relationship between notions of dependence

Theorem 3.7.1. *For a sequence of random variable $(X_0, X_1, X_2, \ldots)$ we have many notions of dependence: iid, exchangeable, stationary and identically distributed. The implications are*

iid sequence $\implies$ exchangeable seq. $\implies$ stationary seq. $\implies$ identically distributed seq.

NB: non fatto questo anno?

Important remark 24. In the special case where (X_n) is a Markov chain, stationarity $\iff$ identically distributed.

iid $\implies$ exchangeability. Every iid sequence is exchangeable. If (X_n) is iid:

$$\begin{aligned}
\mathbb{P}\left(X_{\pi_0} \in A_0, X_{\pi_1} \in A_1, \ldots, X_{\pi_n} \in A_n\right) &\overset{(1)}{=} \prod_{i=0}^{n} \mathbb{P}\left(X_{\pi_i} \in A_i\right) \\
&\overset{(2)}{=} \prod_{i=0}^{n} \mathbb{P}\left(X_i \in A_i\right) \\
&\overset{(3)}{=} \mathbb{P}\left(X_0 \in A_0, X_1 \in A_1, \ldots, X_n \in A_n\right)
\end{aligned}$$

where in

1. (1) due to being independent
2. (2) to being identically distributed
3. (3) using independence again (reversed this time)

Therefore we conclude that

$$(X_{\pi_0}, X_{\pi_1}, \ldots, X_{\pi_n}) \sim (X_0, X_1, \ldots, X_n)$$

and this is the definition of exchangeability. □

Example 3.7.2 (Counterexample exchangeability $\not\Longrightarrow$ iid)**.** For a counterexample (with an exchangeable sequence which is not iid) fix a non degenerate random variable V and define $X_n = V$, forall $n \geq 0$. So the sequence is $(V, V, V, \ldots)$:

- the sequence is trivially exchangeable because if we make any permutation nothing happens

$$(X_{\pi_0}, X_{\pi_1}, \ldots, X_{\pi_n}) = (V, V, \ldots, V)$$

- however the sequence is not independent (nor iid), because the variable is the same and all the values are equal (più dipendente di così non si può)[6]

exchangeable $\implies$ *stationary.* If (X_n) is exchangeable we can take:

$$(\pi_0, \pi_1, \ldots, \pi_n) = (1, 2, \ldots, n, 0)$$

getting that

$$(X_1, X_2, \ldots, X_n, X_0) \sim (X_0, X_1, \ldots, X_{n-1}, X_n)$$

In particular

$$(X_1, X_2, \ldots, X_n) \sim (X_0, X_1, \ldots, X_{n-1}), \quad \forall n \geq 1$$

and this implies

$$(X_1, X_2, X_3, \ldots) \sim (X_0, X_1, X_2, \ldots)$$

so that (X_n) is stationary (and exchangeability $\implies$ stationarity). □

Example 3.7.3 (Counterexample stationary $\not\Longrightarrow$ exchangeable)**.** Consider a symmetric random variable V, that is $V \sim -V$ (eg normal with mean 0), with $\mathbb{P}(V = 0) < 1$ (non degenerate on the centre). Define:

$$X_n = \begin{cases} V & \text{if } n \text{ is even} \\ -V & \text{if } n \text{ is odd} \end{cases}$$

We have to prove that this sequence is stationary but not exchangeable

- to prove it is not exchangeable
$$\mathbb{P}(X_0 = X_2) = \mathbb{P}(V = V) = 1$$
$$\mathbb{P}(X_0 = X_1) = \mathbb{P}(V = -V) \overset{(1)}{=} \mathbb{P}(V = 0) < 1$$
where in (1) $V = -V$ is possible only if $V = 0$. Hence $(X_0, X_1) \nsim (X_0, X_2)$ and thus (X_n) is not exchangeable.

- however (X_n) is stationary: infact, at time 0 we choose the value of V (eg $V = 18$), this is the only random moment; then we go on in a deterministic way (oscillating $18, -18, 18 \ldots$). After we choose V there is nothing

[6] Incidentally recall that given any rv V one obtains

$$V \perp\!\!\!\perp V \iff V \text{ is degenerate}$$

Infact if V is independent of itself then

$$\mathbb{P}(V \in A, V \in B) = \mathbb{P}(V \in A) \cdot \mathbb{P}(V \in B), \forall A, B \in \beta(\mathbb{R})$$

Taking $A = B$

$$\mathbb{P}(V \in A, V \in A) = \mathbb{P}(V \in A)^2 \implies \mathbb{P}(V \in A) \in \{0, 1\}, \quad \forall A \in \beta(\mathbb{R})$$

Hence $\mathbb{P}(V \in A) \in \{0, 1\}, \forall A \in \beta(\mathbb{R})$ and this easily implies that V is degenerate random. So considering both the original and the shifted distribution we have for each $n \geq 1$

$$(X_0 X_1, \dots, X_n) = (V, -V, V, -V, \dots, (-1)^n V)$$
$$(X_1, X_2, \dots, X_{n+1}) = (-V, V, -V, V, \dots (-1)^{n+1} V)$$

these two sequences have the same distribution because $V = -V$ is symmetrical. hence

$$(X_0, X_1, \dots, X_n) \sim (X_1, X_2, \dots X_{n+1}), \forall n$$

thus (X_n) is stationary

NB: non fatto questo anno?

Stationary $\implies$ *identically distribution.* Supposing that (X_n) is stationary, that is $(X_1, X_2, \dots) \sim (X_0, X_1, \dots)$; by induction one obtains that

$$(X_k X_{k+1}, X_{k+2}, \dots) \sim (X_0, X_1, X_2, \dots)$$

In particular $X_k \sim X_0, \forall k \geq 1$, so the sequence is identically distributed. $\square$

NB: non fatto questo anno?

Example 3.7.4 (Counterexample identically distributed $\not\Longrightarrow$ stationary)**.** Take a sequence (Z_i) iid $\mathrm{N}(0,1)$. we define

$$X_0 \sim \mathrm{N}(0,1), X_0 \perp\!\!\!\perp (Z_i)$$
$$X_n = \frac{\sum_{i=1}^n Z_i}{\sqrt{n}}, \forall n \geq 1$$

Now:

- the sequence is clearly identically distributed: X_0 is normal while $\sum_{i=1}^n Z_i$ is again normal $\sim \mathrm{N}(0,n)$ (variance is the sum because independent n of variance 1). Hence $X_n = \sum_{i=1}^n (Z_i)/\sqrt{n} \sim \mathrm{N}(0,1)$ so the sequence (X_n) is identically distributed (each random variable in the sequence has the same distribution, which is a $\mathrm{N}(0,1)$).
- we want to see that it's not stationary; let's evaluate the covariance

$$\begin{aligned}\mathrm{Cov}(X_1, X_2) &\overset{(1)}{=} \mathbb{E}[X_1 \cdot X_2] = \mathbb{E}\left[Z_1 \frac{Z_1 + Z_2}{\sqrt{2}}\right] = \frac{1}{\sqrt{2}} \mathbb{E}[Z_1(Z_1 + Z_2)] \\ &= \frac{\mathbb{E}\left[Z_1^2\right] + \mathbb{E}[Z_1 Z_2]}{\sqrt{2}} \overset{(2)}{=} \frac{1+0}{\sqrt{2}}\end{aligned}$$

where in (1) we avoid $-\mathbb{E}[X_1]\mathbb{E}[X_2]$ (because both terms are null) and in (2) the covariance between $Z_1 Z_2$ is 0 because of their independence.
Hence X_1 is not independent of X_2: the covariance is not null (if they were independent the covariance would be 0) while $X_0 \perp\!\!\!\perp X_1$ by assumption/definition. Thus,

$$(X_0, X_1) \overset{(1)}{\nsim} (X_1, X_2) \implies (X_n) \text{ is not stationary}$$

in (1) because the first (X_0, X_1) are independent while (X_1, X_2) are not dependent.
If X_n was stationary we could say by definition $(X_1, X_2, X_3, \dots) \sim (X_0, X_1, X_2, \dots)$; in particular the first two elements of the first sequence should have the same distribution $(X_1, X_2) \sim (X_0, X_1)$; in this case they have not the same distribution and thus we can conclude it's not stationary.

3.7.3 De Finetti's regresentation thm

Remark 37. The following theorem is an important characterization of exchangeable sequences.

Theorem 3.7.2 (de Finetti representation theorem). *A sequence (X_n) of real rv is exchangeable if and only if:*

$$\mathbb{P}((X_0, X_1, \dots) \in A) = \mathbb{E}_\nu [\mathbb{P}_\nu [(X_0, X_1, \dots) \in A]], \quad \forall \text{ measurable event } A$$

where $\mathbb{P}_\nu [(X_0, X_1, \dots) \in A]$ denotes the probability of $[(X_0, X_1, \dots) \in A]$ under the assumption that (X_n) is iid and $X_0 \sim \nu$, while $\mathbb{E}_\nu$ denotes expectation with respect to ν.

Important remark 25 (Interpretation). The notation given in the definetti thm

$$\mathbb{P}(\text{something}) = \mathbb{E}_\nu \{\mathbb{P}_\nu(\text{something})\}$$

means that the probability of some events, if the sequence is exchangeable, is binded to the probability $\mathbb{P}_\nu(\text{something})$ (which is the probability that something under the assumption that (X_n) is iid with distribution ν) via expectation on all possible probability function ν.
Thus if (X_n) is exchangeable and we want to evaluate a general probability regarding the sequence $X_0 \in A_0, \dots, X_n \in A_n$, we start by evaluating it under the assumption of it being iid. Then we calculate the mean $\mathbb{E}_\nu$ with respect to all the possible probability distributions ν:

$$\begin{aligned}\mathbb{P}(X_0 \in A_0, \dots, X_n \in A_n) &= \mathbb{E}_\nu [\mathbb{P}_\nu [X_0 \in A_0, \dots, X_n \in A_n]] \\ &\overset{(1)}{=} \mathbb{E}_\nu [\mathbb{P}_\nu(X_0 \in A_0) \cdot \ldots \cdot \mathbb{P}_\nu(X_n \in A_n)] \\ &\overset{(2)}{=} \mathbb{E}_\nu [\nu(A_0) \cdot \ldots \cdot \nu(A_n)]\end{aligned}$$

with (1) because of independence and (2) because of identical distribution

Remark 38. The most important example of exchangeable sequence done during the course is the Polya urn at the end, I guess. It needs the following stuff to be fully developed.
Below we use de Finetti's theoreom to prove that strong law of large number holds for exchangeable sequences as well (not only IID as proved by Kolmogorov). In a nutshell:

- if the sequence is IID the sample mean converges to the common mean by Kolmogorov
- in the more general exchangeable case, the sample mean converges to V which is a generic non degenerate random variable (we don't know what is it, we know it exists).

Proposition 3.7.3 (Convergence of exchangeable sequence mean). *If (X_n) is any exchangeable sequence such that $\mathbb{E}[|X_0|] < +\infty$ (having the mean), then the sample mean sequences converges to a random variable V*

$$\overline{X}_n = \frac{1}{n} \sum_{i=0}^{n-1} X_i \xrightarrow{a.s.} V$$

for some random variable V.

Dimostrazione. If succession was iid with finite mean by Kolmogorov SLLN we had almost sure convergence and probability of convergence would be 1. Here we don't have iid, but being exchangeable and using deFinetti, we can calculate probability of event (sample mean converge), like it was iid.
By de Finetti thm we have that

$$\mathbb{P}\left(\overline{X}_n \text{ converges}\right) = \mathbb{E}_\nu\left(\mathbb{P}_\nu(\overline{X}_n \text{ converges})\right)$$

where $\mathbb{P}_\nu$ denotes the probability distribution of an iid sequence with common distribution ν. So the probability that the sample mean converges is equal to the expectation (with respect to ν) of the probability of convergence, under assumption of iid sequence with common distribution ν.
But in the hypothesis the sequence is iid and the mean exists, the probability of convergence is 1 by strong law of large numbers of Kolmogorov so

$$\mathbb{P}\left(\overline{X}_n \text{ converges}\right) = \mathbb{E}_\nu\left(\mathbb{P}_\nu(\overline{X}_n \text{ converges})\right) = E_\nu(1) = 1$$

This proves that if the sequence (X_n) is exchangeable and the mean exists ($\mathbb{E}\left[|X_0|\right] < +\infty$) then the sample mean $\overline{X} \xrightarrow{a.s.} V$.
This years' proof follows. Consider the event:

$$H = \left\{\omega \in \Omega : \text{the sequence } \left(\overline{X}_n(\omega)\right) \text{ converges to a finite limit}\right\}$$

if $\mathbb{P}\left(H\right) = 1$ to prove the SLLN, it suffices to let/assume

$$V(\omega) = \begin{cases} \lim_{n\to+\infty} \overline{X}_n(\omega) & \text{if } \omega \in H \\ V(\omega) = 0 & \text{if } \omega \notin H \end{cases}$$

And, to show that $P(H) = 1$, it suffices to apply de Finetti's thm

$$P(H) \overset{(1)}{=} \mathbb{E}_\nu\left[\mathbb{P}_\nu(H)\right] \overset{(2)}{=} \mathbb{E}_\nu(1) = 1$$

where in

- (1) since (X_n) is exchangeable;
- (2) $\mathbb{P}_\nu(H)$ is the probability of H under the assumption that (X_n) is iid with common distribution ν. Hence $\mathbb{P}_\nu(H) = 1$ follows from Kolmogorov's SLLN

□

Important remark 26 (Some truth bomb dropped)*.* To appreciate exchangeability think of statistical inference. Suppose we aim to learn something on a unknown parameter $\theta \in \Theta$:

- in the classical approach, θ is regarded as an unknown constant and the usual model is

  $\forall\theta \in \Theta$the data are iid according to a common distribution depending on θ

  For istance $\Theta = \mathbb{R}$ and $\forall\theta \in \mathbb{R}$ the data are iid with common distribution $\mathrm{N}\left(\theta, 1\right)$

- on the contrary, in the bayesian approach, θ is regarded as a random variable and the usual model is "the data are *conditionally* iid, given the rv θ" or equivalently "the data are exchangeable".
  Practically, the Bayesian inferer has to select the distribution of θ, teh so called *prior* distribution, and then has to calculate the conditional distribution of θ given the data, the so called *posterior* distribution.
  Note that, since the parameter is regarded as rv (and not as unknown constant) to estimate it does not make sense. All one can do is to evaluate the conditional distribution of θ given the data, namely the posterior distribution

Example 3.7.5 (An exchageable sequence)**.** Let (Y_n) be an iid sequence and $Z \perp\!\!\!\perp (Y_n)$

NB: non fatto, ergo non rivisto

$$X_n = Z + Y_n, \forall n \geq 0$$

(X_n) is not iid because all the X_n shares a common factor Z so can't be independent. However X_n is exchangeable because we can write:

$$\begin{aligned}
\mathbb{P}\left(X_0 \in A_0, \ldots, X_n \in A_n\right) &\overset{(0)}{=} \mathbb{E}_z\left[\mathbb{P}\left(X_0 \in A_0, \ldots, X_n \in A_n | Z = z\right)\right] \\
&= \mathbb{E}_z\left[\mathbb{P}\left(Z + Y_0 \in A_0, \ldots, Z + Y_n \in A_n | Z = z\right)\right] \\
&= \mathbb{E}_z\left[\mathbb{P}\left(z + Y_0 \in A_0, \ldots, z + Y_n \in A_n | Z = z\right)\right] \\
&\overset{(1)}{=} \mathbb{E}_z\left[\mathbb{P}\left(z + Y_0 \in A_0, \ldots, z + Y_n \in A_n\right)\right] \\
&\overset{(2)}{=} \mathbb{E}_z\left[\mathbb{P}\left(z + Y_0 \in A_0\right) \cdot \ldots \cdot \mathbb{P}\left(z + Y_n \in A_n\right)\right] \\
&\overset{(3)}{=} \mathbb{E}_z\left[\mathbb{P}\left(z + Y_0 \in A_0\right) \cdot \ldots \cdot \mathbb{P}\left(z + Y_0 \in A_n\right)\right]
\end{aligned}$$

where in

- (0) in order to prove the sequence is exchangeable, in a sense in the first step we need to regard Z as a constant; Z is not a constant, it's a rv, but if we condition on Z it becomes a constant. In probability every time we want to treat something as fixed for some reason we can condition on its values. Here we treat like a constant because in the case the sequence $z + Y_n$ becomes iid;
- (1) since $Z \perp\!\!\!\perp Y$ i drop conditioning;
- (2) the the sequence $(z + Y_n)$ is iid for every fixed value of z, and thus probability of intersection is product of marginals
- (3) being iid.

Thus (X_n) is exchangeable by de Finetti theorem (ci si poteva fermare alla penultima per vederlo direi).
Idea of the example: Y_n is iid (and thus exchangeable); Z is independent of Y_n; if I sum them exchangeability is preserved but not independence because all the sums have in common Z. The sequence $(Z + Y_n)$ is not iid, while $(z + Y_n)$ is iid for each fixed z. at the last passage it remains to make the expectation with respect to Z but this depends on the distribution of Z. however, whatever distribution of Z, we have exchangeable because $z + Y_0, z + Y_1, \ldots$ is iid and thus the probability of our interest depends on Z but the probability is invariant under permutation. And the conclusion is that X_n is exchangeable

Example 3.7.6. Suppose we have an urn with black and white balls, drawing with replacement, $\theta \in (0,1)$ be the proportion of white ball in the urn. Let X_n be the indicator

NB: non fatto quest'a

$$X_n = \begin{cases} 1 & \text{white ball at trial } n \\ 0 & \text{otherwise} \end{cases}$$

We're making drawings with replacement so, supposing we want to evaluate the general event $\mathbb{P}(X_1 = x_1, \dots, X_n = x_n)$, considered that $x_1, x_2, \dots x_n \in \{0,1\}$ and the sequence is iid due to replacement:

$$\begin{aligned} \mathbb{P}(X_1 = x_1, \dots, X_n = x_n) &= \mathbb{P}(X_1 = x_1) \cdot \dots \cdot \mathbb{P}(X_n = x_n) \\ &= \theta^{\text{n. of white balls}} \cdot (1-\theta)^{\text{n. of black balls}} \\ &= \theta^{\sum_{i=1}^n x_i} (1-\theta)^{n - \sum_{i=1}^n x_i} \end{aligned}$$

So far exchangeability doesn't have a role.
However suppose that θ is unknown. We need it to estimate our probability:

- we could estimate θ, considered as unknown constant/parameter, by maximum likelihood in a frequentist approach
- on the other way going bayesian we treat θ like a random variable (don't knowing it) so having a distribution object of interest.

TODO: non chiaro perché qui non usa il fatto che iid implica scambiabile e lo assume

In the second case it seems reasonable to assume it's exchangeable, thus we can apply de Finetti's theorem. Assuming there's a distribution of θ (in this case the ν of the theorem statement is θ) and we take the expected value to evaluate the probability calculated only under iid hypothesis so:

$$\mathbb{P}(X_1 = x_1, \dots, X_n = x_n) = \mathbb{E}_\theta \left[\theta^{\sum_{i=1}^n x_i} (1-\theta)^{\sum_{i=1}^n x_i} \right]$$

where $\mathbb{E}_\theta$ means the expectation with respect to θ.
For instance if θ has absolutely continuous distribution with density f (which must be limited to $[0,1]$ being θ a probability), then

$$\begin{aligned} \mathbb{P}(X_1 = x_1, \dots, X_n = x_n) &= \mathbb{E}_\theta \left[\theta^{\sum_{i=1}^n x_i} (1-\theta)^{\sum_{i=1}^n x_i} \right] \\ &= \int_0^1 \left(\theta^{\sum_{i=1}^n x_i} \cdot (1-\theta)^{\sum_{i=1}^n x_i} \right) \cdot f(\theta) \, \mathrm{d}\theta \end{aligned}$$

In this case distribution of θ, $f(\theta)$, is the so called prior.
So assuming the sequence is exchangeable and by assigning a proper prior distribution of θ I can evaluate any event regarding the sequence.

Example 3.7.7 (Polya urn)**.** Instead of drawing with or witout replacement another extraction scheme is dued to Polya.
We have an urn with $a > 0$ white and $b > 0$ black ball: at each time i draw a ball, look at its color and put it back in the urn adding k more balls of the same color. Let X_n be the indicator:

$$X_n = \begin{cases} 1 & \text{white ball at trial } n \\ 0 & \text{black ball at trial } n \end{cases}$$

At

- first extraction probability of white ball is

$$\mathbb{P}(X_0 = 1) = \frac{a}{a+b}$$

- second extraction the conditional probability of white ball is:

$$\mathbb{P}(X_1 = 1|X_0 = x) = \frac{a+kx}{a+b+k}$$

  where:

  - at the denominator the total number of balls at second extraction: $a+b+k$
  - at the numerator the number of white depends on first extraction. If $X_0 = x = 1$, we have $a+k$, while if $x = 0$, a, so we can summarize by using x.

- n-th extraction, in general, the conditional probability of white ball given previous extraction with $x_0, \ldots, x_{n-1} \in \{0, 1\}$

$$\mathbb{P}(X_n = 1|X_0 = x_0, \ldots, X_{n-1} = x_{n-1}) = \frac{a + k\sum_{i=0}^{n-1} x_i}{a+b+kn}$$

  while the probability of a black ball is

$$\mathbb{P}(X_n = 0|X_0 = x_0, \ldots, X_{n-1} = x_{n-1}) = \frac{b + k\left(n - \sum_{i=0}^{n-1} x_i\right)}{a+b+kn}$$

Now we see that thanks this conditional probability, the sequence is exchangeable. Suppose we want to evaluate the probability of a certain sequence, we can progressively factor the joint as marginals times conditionals [7]:

$$\begin{aligned}
\mathbb{P}(X_0 = x_0, \ldots, X_n = x_n) &= \mathbb{P}(X_0 = x_0, \ldots, X_{n-1} = x_{n-1}) \cdot \mathbb{P}(X_n = x_n|X_0 = x_0, \ldots, X_{n-1} = x_{n-1}) \\
&= \mathbb{P}(X_0 = x_0, \ldots, X_{n-2} = x_{n-2}) \cdot \mathbb{P}(X_{n-1} = x_{n-1}|X_0 = x_0, \ldots, X_{n-2} = x_{n-2}) \cdot \\
&\quad \cdot \mathbb{P}(X_n = x_n|X_0 = x_0, \ldots, X_{n-1} = x_{n-1}) \\
&= \mathbb{P}(X_0 = x_0) \cdot \prod_{i=0}^{n-1} \mathbb{P}(X_{i+1} = x_{i+1}|X_0 = x_0, \ldots, X_i = x_i)
\end{aligned}$$

Thing is that each factor of the above product has a known form (developed previously); by replacing these factors with their expressions one obtains

$$\begin{aligned}
&\mathbb{P}(X_0 = x_0, \ldots, X_n = x_n) \\
&\quad = \frac{[a+k(j-1)]\,[a+k(j-2)] \cdot \ldots \cdot [a+k] \cdot a \cdot [b+k(n-j)]\,[b+k(n-j-1)] \cdot \ldots \cdot [b+k] \cdot b}{[a+b+k \cdot n] \cdot [a+b+k \cdot (n-1)] \cdot \ldots \cdot [a+b+k] \cdot [a+b]}
\end{aligned}$$

[7] A simple example

$$\begin{aligned}
\mathbb{P}(X_0 = 0, X_1 = 1, X_2 = 1) &= \mathbb{P}(X_0 = 0, X_1 = 1) \cdot \mathbb{P}(X_2 = 1|X_0 = 0, X_1 = 1) \\
&= \mathbb{P}(X_0 = 0) \cdot \mathbb{P}(X_1 = 1|X_0 = 0) \cdot \mathbb{P}(X_2 = 1|X_0 = 0, X_1 = 1) \\
&= \frac{b}{a+b} \cdot \frac{a}{a+b+k} \cdot \frac{a+k \cdot 1}{a+b+2k}
\end{aligned}$$

depends only on a, b and j (number of white), not on x_0, x_1.

where $j = \sum_0^n x_i$ is the number of times we extracted a white balls in trials from 0 to n.
We note that $\mathbb{P}(X_0 = x_0, \dots, X_n = x_n)$ depends on $X_0, \dots, X_n$ only through the sum $j = \sum_0^n x_i$ and not on the single values ($x_0, \dots, x_n$ are not in the final formula, which apart from j depends on a, b, k, n, which are given).
This implies that if I make a permutation the sum j remains the same and the generated probability as well, so

$$\mathbb{P}(X_0 = x_0, \dots, X_n = x_n) = \mathbb{P}(X_{\pi_0} = x_0, \dots, X_{\pi_n} = x_n)$$

This holds $\forall$ permutations $(\pi_0, \dots, \pi_n)$ of $(0, \dots, n)$. Hence the sequence (X_n) is exchangeable

NB: non fatta

Example 3.7.8 (Polya part 2)**.** Being exchangeable, by de Finetti's thm we can write the probability of interest as product of iid events, and taking the expected value for all the possible value of the unknown parameter:

$$\mathbb{P}(X_0 = x_0, \dots, X_n = x_n) = \mathbb{E}_\theta\left[\theta^{\sum_0^n x_i} \cdot (1-\theta)^{\sum_0^n (1-x_i)}\right]$$

In the special case of Polya urn we can also find the distribution of θ. To do that we need 3.7.3
Let us go back to Polya sequences/urn; by what already proved $\overline{X} \xrightarrow{a.s.} V$. Moreover, given an integer $m \geq 1$, convergence is also in L_m (converge in L_m perché la successione è limitata, le medie campionarie sono tutte ≤ 1, e posso maggiorare di una costante il valore assoluto; questo basta per dire che la convergenza a.s. implica la convergenza in LM (cosa che normalmente non avviene)..
Hence:

$$\begin{aligned}
\mathbb{E}[V^m] &= \lim_n \mathbb{E}\left[\overline{X}_n^m\right] = \lim_n \mathbb{E}\left[\left(\frac{\sum_{i=0}^{n-1} X_i}{n}\right)^m\right] = \lim_n \frac{1}{n^m}\mathbb{E}\left[\left(\sum_0^{n-1} X_i\right)^m\right] \\
&= \lim_n \frac{1}{n^m} \sum_{j_1=0}^{n-1}\sum_{j_2=0}^{n-1}\dots\sum_{j_m=0}^{n-1} \mathbb{E}[X_{j_1}\cdot \dots \cdot X_{j_m}] \\
&\overset{(1)}{=} \lim_n \frac{1}{n^m}\left[\underbrace{\sum_{j_1=0}^{n-1}\sum_{j_2=0}^{n-1}\dots\sum_{j_m=0}^{n-1}}_{j_1,\dots,j_m \text{ distinct}} \mathbb{E}[X_{j_1}\cdot \dots \cdot X_{j_m}] + \underbrace{\sum_{j_1=0}^{n-1}\sum_{j_2=0}^{n-1}\dots\sum_{j_m=0}^{n-1}}_{j_1,\dots,j_m \text{ not distinct}} \mathbb{E}[X_{j_1}\cdot \dots \cdot X_{j_m}]\right] \\
&\overset{(2)}{=} \lim_n \frac{1}{n^m}\left[n(n-1)\dots(n-m+1)\,\mathbb{E}[X_0X_1,\dots X_{n-1}] + \underbrace{\sum_{j_1=0}^{n-1}\sum_{j_2=0}^{n-1}\dots\sum_{j_m=0}^{n-1}}_{j_1,\dots,j_m \text{ not distinct}} \mathbb{E}[X_{j_1}\cdot \dots \cdot X_{j_m}]\right]
\end{aligned}$$

where in:

- (1) we split the sum in two parts where in the first all the indexes are all distinct, while in the second there are repetitions

- in (2) where $j_1, \ldots, j_m$ are all distinct under exchangeability the distribution is invariant under permutation so
$$\mathbb{E}\left[X_{j_1} \cdot \ldots \cdot X_{j_m}\right] = \mathbb{E}\left[X_0 \cdot \ldots \cdot X_{n-1}\right]$$
and we have $n(n-1) \cdot \ldots \cdot (n-m+1)$ of these expected values with different indexes (n choices as first index, ..., $(n-m+1)$ as the m-th index)

Now we have that the two terms goes like
$$\frac{n(n-1) \cdot \ldots \cdot (n-m+1)}{n^m} \to 1 \quad \text{as } n \to \infty$$
$$\frac{\sum_{j_1} \sum_{j_2} \ldots \sum_{j_m} \mathbb{E}\left[X_{j_1} \cdot \ldots \cdot X_{j_m}\right]}{n^m} \to 0$$

Regarding the second factor we don't know the value of the expectation $\mathbb{E}\left[X_{j_1} \cdot \ldots \cdot X_{j_n}\right]$ but surely is ≤ 1 (since each X can be only 0 or 1) so we can upper bound it to one and just use the cardinality of the double sum:
$$\frac{\overbrace{\sum_{j_1} \ldots \sum_{j_m}}^{\text{not all distinct}} \mathbb{E}\left[X_{j_1} \ldots X_{j_m}\right]}{n^m} \overset{(1)}{\leq} \frac{n^m - n(n-1)\ldots(n-m+1)}{n^m} = 1 - \frac{n(n-1)\ldots(n-m+1)}{n^m} \to 1 - 1 = 0$$
where in (1) being $n(n-1) \cdot \ldots \cdot (n-m+1)$ the number of distict touples and n^m all the touples, $n^m - n(n-1) \cdot \ldots \cdot (n-m+1)$ are the touples with at least a repeated index.
Hence
$$\begin{aligned}
\mathbb{E}\left[V^m\right] &= \mathbb{E}\left[X_0 X_1 \ldots X_{n-1}\right] \\
&\overset{(1)}{=} \mathbb{P}\left(X_0 = X_1 = \ldots = X_{n-1} = 1\right) \\
&\overset{(2)}{=} \frac{[a+k(m-1)][a+k(m-2)] \cdot \ldots \cdot a}{[a+b+k(m-1)] \cdot \ldots \cdot [a+b]} \\
&\overset{(3)}{=} \frac{(\frac{a}{k} + (m-1))(\frac{a}{k} + (m-2)) \cdot \ldots \cdot \frac{a}{k}}{(\frac{a+b}{k} + (m-1))(\frac{a+b}{k} + (m-2)) \cdot \ldots \cdot \frac{a+b}{k}}
\end{aligned}$$
where in:

- (1) since X_i are indicators the product of indicators is still an indicator (all $= 1$), and the expected value of indicator is the probability of its event;
- (2) we apply the formula derived before;
- (3) we rewrited by dividing by k.

The last equation is the moment of order m of a beta random variable with parameters $\frac{a}{k}$ and $\frac{b}{k}$.
Now, in general $\mathbb{E}\left[X^m\right] = \mathbb{E}\left[Y^m\right], \forall m \;\not\Longrightarrow\; X \sim Y$. However if furthermore X (or Y) has finite moment generating function the implication holds.
Here since $0 \leq V \leq 1$, then V has finite moment generating function: if $a \leq V \leq b$ we have that
$$e^{at} \leq \mathbb{E}\left[e^{tV}\right] \leq e^{bt}$$

and the moment generating function $\mathbb{E}\left[e^{tV}\right]$ is finite.
Therefore having V the same moments of a beta and finite moment generating function, then V is beta distributed $V \sim \text{Beta}\left(\frac{a}{k}, \frac{b}{k}\right)$.
Hence we can say that $\theta = V$ is beta distributed and finally

$$\begin{aligned}\mathbb{P}\left(X_0 = x_0, \dots, X_n = x_n\right) &= \mathbb{E}_\theta\left[\theta^{\sum_0^n x_i}(1-\theta)^{\sum_0^n (1-x_i)}\right] \\ &= \int_0^1 \theta^{\sum_0^n x_i}(1-\theta)^{\sum_0^n (1-x_i)} \cdot f(\theta)\, \mathrm{d}\theta\end{aligned}$$

where f is a density of a beta distribution with parameter $V \sim \text{Beta}\left(\frac{a}{k}, \frac{b}{k}\right)$.

Capitolo 4

Brownian motions

Remark 39. So far we've seen discrete time process (Martingales and Markov chains); now we focus on continuous time random processes. Brownian motions are the first of this kind.

4.1 Introduction to continuous time process

Definition 4.1.1 (Continuos random processes (recap))**.** A process X where the random variables are indexed by a set T, interval on the real line (eg $T = [0, +\infty)$). We consider

$$X = \{X_t : t \geq 0\}$$

as our main process.

Definition 4.1.2 (Filtration)**.** It is now a collection of σ-fields $(\mathscr{F}_t : t \geq 0)$ such that $\mathscr{F}_s \subset \mathscr{F}_t \subset \mathscr{A}$, $\forall 0 \leq s < t$.

Remark 40. Interpretation is the same: $\mathscr{F}_t$ is our information at time t. In most situations is reasonable to assume that the information increases.

Definition 4.1.3 (Stopping time)**.** It's a function $\tau : \Omega \to [0, +\infty]$ such that the set $\{\tau \leq t\} \in \mathscr{F}_t$, $\forall t \geq 0$.

Remark 41. Interpretation is the same: time when something of interest happens (if $+\infty$ the event we're waitinig for doesn't happen)

Definition 4.1.4 (Martingale as cont. time process)**.** X is a martingale if

NB: non fatto quest'anno

1. $\mathbb{E}[|X_t|] < +\infty$, $\forall t \geq 0$
2. X is adapted
3. $\mathbb{E}[X_t|\mathscr{F}_s] = X_s$, $\forall 0 \leq s < t$

Remark 42. That is the notion extend naturally/the same way. Now we're ready for the definition of interest

4.2 Brownian motion

Definition 4.2.1 (Brownian motion)**.** Let $X = \{X_t : t \geq 0\}$ bea continuous process and $(\mathscr{F}_t)$ a filtratio. X is a brownian motion (BM) with respect to $(\mathscr{F}_t)$ if:

1. X is adapted to $(\mathscr{F}_t)$, namely X_t is $\mathscr{F}_t$-measurable, $\forall t \geq 0$
2. X has *indipendent increments*
3. $X_0 = 0$ almost surely and $X_t \sim \mathrm{N}\left(0, \sigma^2 t\right)$, $\forall t > 0$, where $\sigma^2 > 0$ is a parameter

If $\sigma^2 = 1$, X is called standard brownian motion (SBM)

Definition 4.2.2 (Process with independent increments)**.** A process $X = \{X_t : t \geq 0\}$ is said to have independent increments if

$$(X_t - X_s) \perp\!\!\!\perp \mathscr{F}_s, \forall s, t : 0 \leq s < t$$

Remark 43. Heuristically, this means that the random variation $X_t - X_s$ between s and t (with $s < t$) is independent on the information up to time s (described by the σ-field $\mathscr{F}_s$).
If X is adapted, the information up to time s includes the *path* (pensa alla serie storica) of X on the interval $[0, s]$. In this case, in particular, $X_t - X_s$ is independent of the path of X on $[0, s]$.
So at time s we can predict the increment ignoring all the stuff before now.

Remark 44 (History). During '800 Brown, a botanist, describe the motion of pollen in a fluid; 30 years after, Wiener describe a mathematical model for that motion. For this reason brownian motion can be called Wiener process as well.
The so called *Wiener process* is a triplet $(X_t, Y_t, Z_t) : t \geq 0$, where X, Y, Z are independent brownian motion (we need 3 coordinates to describe the motion in the space)

4.3 Properties/features of BM

Remark 45. Now we see some features of BMs. Before that a definition.

Definition 4.3.1 (Property satisfied up to equivalence)**.** A process $X = \{X_t : t \in T\}$ is said to satisfy a property up to equivalence if there is *another* process $Z = \{Z_t : t \in T\}$ such that:

1. Z satisfies the property
2. $\mathbb{P}(X_t \neq Z_t) = 0, \forall t \in T$ (Z is "close" to X, in this sense)

Remark 46 (Interpretation). Hence the property is satisfied up to equivalence either

- if the process X satisfies it itself (in which case one can take $Z = X$) *or*
- it may be that X does *not* satisfy the property but there's another process which satisfies it and is very close to X (in this case we can "throw away" X and use Z).

4.3.1 Erratic paths

Exam question: talk t the path of brow- motions (then we say hings below)

Important remark 27 (Paths of a brownian motion and their main properties). Paths $X(t, \omega)$ of brownian motions are functions (fixed $\omega \in \Omega$) in t from $[0, +\infty) \to \mathbb{R}$ which are

1. *continuous* and
2. and *nowhere differentiable*[1] ...

... *up to equivalence.*

Remark 47. So

- it's something extremely erratic, hard to make a picture.
- if not X can't be said continuous and nowhere differentiable, there's exists a process very similar to X which is.

4.3.2 Has stationary increments

Definition 4.3.2 (Process with stationary increments)**.** A process X having *independent increments* is said to have *stationary increments* if the distribution of the increment $(X_t - X_s)$ is invariant under shifts of time which means that

$$(X_t - X_s) \sim (X_{t-s} - X_0), \forall 0 \leq s < t$$

That is the distribution of the increment $(X_t - X_s)$ depends only on $t - s$

Proposition 4.3.1 (Stationarity of increments of a BM)**.** *The BM has* stationary increments*; hence since* $X_0 = 0$ *almost surely one obtain*

$$X_t - X_s \sim X_{t-s} \sim \mathrm{N}\left(0, \sigma^2(t-s)\right)$$

Dimostrazione. As a consequence of the definition, a BM has stationary increments. Fix $0 \leq s < t$ and note that

$$X_t = X_s + (X_t - X_s)$$

with $X_s \perp\!\!\!\perp (X_t - X_s)$ because of independent increments.
Hence using characteristic functions, one obtains

$$\varphi_{X_t}(u) = \varphi_{X_s}(u) \cdot \varphi_{X_t - X_s}(u) \forall u \in \mathbb{R}$$

but since $X_t \sim \mathrm{N}\left(0, \sigma^2 t\right)$ and $X_s \sim \mathrm{N}\left(0, \sigma^2 s\right)$ their characteristic functions are known and one can obtain the characteristic function of $X_t - X_s$ as

$$\varphi_{X_t - X_s}(u) = \frac{\varphi_{X_t}(u)}{\varphi_{X_s}(u)} = \frac{e^{-\frac{1}{2}u^2\sigma^2 t}}{e^{-\frac{1}{2}u^2\sigma^2 s}}$$
$$= e^{-\frac{1}{2}u^2\sigma^2(t-s)}$$

which is the characteristic function of $\mathrm{N}\left(0, \sigma^2(t-s)\right)$.
Since $X_0 = 0$ a.s. and $X_t - X_s \sim \mathrm{N}\left(0, \sigma^2(t-s)\right)$ one obtains

$$X_t - X_s \sim X_{t-s} = X_{t-s} - X_0$$

Hence X has stationary increments. □

[1] A function $f : I \to \mathbb{R}$ is said to be nowhere differentiable if, $\forall t \in I$, f is not differentiable at t)

4.3.3 It's a martingale

Proposition 4.3.2. *If $X = \{X_t : t \in T\}$ is brownian motion then is a martingale.* [2]

Dimostrazione. In this course when we want to prove that something is a martingale we prove only the third properties of their definition (taking the first two as given). To prove

$$\mathbb{E}\left[X_t|\mathscr{F}_s\right] = X_s, \quad \forall 0 \le s < t$$

fix s with $0 \le s < t$ then

$$\begin{aligned}
\mathbb{E}\left[X_t|\mathscr{F}_s\right] &= \mathbb{E}\left[X_s + (X_t - X_s)|\mathscr{F}_s\right] = \mathbb{E}\left[X_s|\mathscr{F}_s\right] + \mathbb{E}\left[X_t - X_s|\mathscr{F}_s\right] \\
&\overset{(1)}{=} X_s + \mathbb{E}\left[X_t - X_s|\mathscr{F}_s\right] \overset{(2)}{=} X_s + \mathbb{E}\left[X_t - X_s\right] \\
&= X_s + (\underbrace{\mathbb{E}\left[X_t\right] - \mathbb{E}\left[X_s\right]}_{=0}) = X_s
\end{aligned}$$

where in

- (1) since it's adapted
- (2) since process has independent increments $X_t - X_s \perp\!\!\!\perp \mathscr{F}_s$ so we drop conditioning

□

4.3.4 Covariance between terms

Proposition 4.3.3 (Covariance between terms of a BM)**.** *We have:*

$$\operatorname{Cov}\left(X_s, X_t\right) = \sigma^2 \cdot (s \wedge t)$$

Dimostrazione. Suppose $0 \le s \le t$: then

$$\begin{aligned}
\operatorname{Cov}\left(X_s, X_t\right) &= \mathbb{E}\left[X_s X_t\right] - \mathbb{E}\left[X_s\right]\mathbb{E}\left[X_t\right] \overset{(1)}{=} \mathbb{E}\left[X_s X_t\right] \\
&= \mathbb{E}\left[X_s(X_s + X_t - X_s)\right] \\
&\overset{(2)}{=} \underbrace{\mathbb{E}\left[X_s^2\right]}_{\sigma^2 s} + \underbrace{\mathbb{E}\left[X_s(X_t - X_s)\right]}_{0} \\
&= \sigma^2 s
\end{aligned}$$

- where in (1) both $\mathbb{E}\left[X_s\right], \mathbb{E}\left[X_t\right]$ are null;
- in (2) since X_s has null expectation, the $\mathbb{E}\left[X_s^2\right]$ is the variance, and by definition of brownian motion is $\sigma^2 s$. On the other hand, the covariance between X_s and the increment is zero for independence increment (the covariance would actually be $\mathbb{E}\left[X_s(X_t - X_s)\right] - \mathbb{E}\left[X_s\right]\mathbb{E}\left[X_t - X_s\right]$ but the last part is null);

[2] Versione estesa non fatta quest'anno: Let $Y_t = X_t^2 - \sigma^2 t$. If X is brownian motion then X and Y are both martingales. Proof is commented.

So if $s \leq t$

$$\text{Cov}\,(X_s, X_t) = \sigma^2 s$$

but more in general

$$\text{Cov}\,(X_s, X_t) = \sigma^2 (s \wedge t)$$

$\square$

4.3.5 Is a gaussian process

Definition 4.3.3 (Gaussian process)**.** A process $X = \{X_t : t \in T\}$ is said to be gaussian if all finite dimension distribution are normal that is

$$(X_{t_1}, \dots, X_{t_n}) \sim N, \quad \forall n \geq 1, \forall t_1, \dots, t_n \in T$$

Remark 48. So it's gaussian if all the finite dimensional distribution are normal. Gaussian means not only $X \sim N$, $\forall t$: even bivariate and multivariate random vector should be normal.
Again, be careful: if X is any process, it may be that $X_t \sim \text{N}, \forall t$ but $(X_s, X_t) \nsim \text{N}$ for some (s, t); in this case X is not gaussian

Proposition 4.3.4. *A BM is a gaussian process, in the sense that*

$$\begin{bmatrix} X_{t_1} \\ X_{t_2} \\ \dots \\ X_{t_n} \end{bmatrix} \sim \text{N}, \quad \forall \textit{ finite collection } \{t_1, \dots, t_n\} \textit{ of time points}$$

Dimostrazione. Fixed $0 \leq t_1 \leq t_2 \leq t_n$ the vector composed by first time and subsequent deltas is normally distributed

$$\begin{bmatrix} X_{t_1} \\ X_{t_2} - X_{t_1} \\ \dots \\ X_{t_n} - X_{t_{n1}} \end{bmatrix} \sim \text{MVN} \left(\mathbf{0}, \begin{bmatrix} \sigma^2 t_1 & 0 & \dots & 0 \\ 0 & \sigma^2(t_2 - t_1) & \dots & 0 \\ \dots & & & \\ 0 & 0 & \dots & \sigma^2(t_n - t_{n-1}) \end{bmatrix} \right)$$

Infact, each single rv is normal, namely $X_{t_1} \sim \text{N}, X_{t_2} - X_{t_1} \sim \text{N}, \dots X_{t_n} - X_{t_{n-1}} \sim \text{N}$ and moreover such rvs are independent. It follows that we can write the subvector of rvs at selected times as linear combination of this vector (and thus still multivariate normal)

$$\begin{bmatrix} X_{t_1} \\ X_{t_2} \\ \dots X_{t_n} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ 1 & 1 & 0 & \dots & 0 \\ 1 & 1 & 1 & \dots & 0 \\ \dots & & & & \\ 1 & 1 & 1 & \dots & 1 \end{bmatrix} \begin{bmatrix} X_{t_1} \\ X_{t_2} - X_{t_1} \\ X_{t_3} - X_{t_2} \\ \dots \\ X_{t_n} - X_{t_{n1}} \end{bmatrix}$$

$\square$

4.4 Alternative definitions

Remark 49. Above we've presented the official (most important/popular) definition of brownian motion: however there are many other equivalent definitions. We see three more of them.
A drawback of these definitions is that they refer to/work with the *canonical* filtration (while the definition above refers to an arbitrary filtration $(\mathscr{F}_t)$).

Definition 4.4.1 (Canonical filtration of X)**.** It's defined as the least σ-field which makes X_s measurable/X adapted, $\forall s \leq t$

$$(\mathscr{F}_t^*) = \sigma(X_s : s \leq t)$$

Supposing we work with this filtration, the only information we have at time t is the path of the process up to time t.

Definition 4.4.2 (Alternative definitions)**.** Let X be any process such that $X_0 = 0$ almost surely. Then X is brownian motion with respect to $(\mathscr{F}_t^*)$ if and only iff (any of the following equivalent situation occurs[3]):

1. X is gaussian, $\mathbb{E}\left[X_t\right] = 0$, $\text{Cov}\left(X_s, X_t\right) = \sigma^2(s \wedge t)$, $\forall s, t$
2. X has stationary independent increments and continuous paths up to equivalence
3. X and $Y = \left\{Y_t = X_t^2 - \sigma^2 t : t \in T\right\}$ are both martingale and X has continuous paths up to equivalence

4.4.1 All on SBM

Example 4.4.1 (A transformation of SBM which is still SBM)**.** Considered a *standard* brownian motion X (such as $\sigma^2 = 1$) we prove that the process Y, defined as

$$\begin{aligned} &Y_0 = 0, \\ &Y_t = tX_{\frac{1}{t}}, \quad \forall t > 0 \end{aligned}$$

is still a standard brownian motion. To prove it we could use any of the alternative definitions of BM, but here the first is convenient. For the various parts of first definition:

- it is gaussian: infact we can write Y as a linear transformation of normally distributed vector

$$\begin{bmatrix} Y_{t_1} \\ \dots \\ Y_{t_n} \end{bmatrix} = \begin{bmatrix} t_1 & 0 & \dots & 0 \\ 0 & t_2 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & t_n \end{bmatrix} \underbrace{\begin{bmatrix} X_{\frac{1}{t_1}} \\ X_{\frac{1}{t_2}} \\ \dots \\ X_{\frac{1}{t_n}} \end{bmatrix}}_{\text{normally distr.}}$$

  So being Y a linear transformation of a normally distributed vector, any linear transformation of normal is normal;

[3]The three definitions are in $\iff$ each other

- turning to the expected value

$$\mathbb{E}[Y_t] = \mathbb{E}\left[tX_{\frac{1}{t}}\right] = t\,\underbrace{\mathbb{E}\left[X_{\frac{1}{t}}\right]}_{=0} \overset{(1)}{=} 0$$

  where in (1) the expected value is 0 because of assumptions (standard brownian motion).

- finally regarding the covariance

$$\begin{aligned}
\operatorname{Cov}(Y_s, Y_t) &= \mathbb{E}[Y_s Y_t] - \mathbb{E}[Y_s]\,\mathbb{E}[Y_t] = \mathbb{E}[Y_s Y_t] \\
&= \mathbb{E}\left[stX_{\frac{1}{s}}X_{\frac{1}{t}}\right] = st\,\mathbb{E}\left[X_{\frac{1}{s}}X_{\frac{1}{t}}\right] \overset{(1)}{=} st\left(\frac{1}{s} \wedge \frac{1}{t}\right) \\
&\overset{(2)}{=} \frac{st}{\max(s,t)} \overset{(3)}{=} \frac{\max(s,t)\min(s,t)}{\max(s,t)} = \min(s,t) \\
&= s \wedge t = (s \wedge t) \cdot 1 =
\end{aligned}$$

  where:

  - in (1) being a standard brownian motion, the covariance is the minimum between $\frac{1}{s}$ and $\frac{1}{t}$;
  - in (2) chossing the minimum between ratio is like choosing the maximum at the denominator;
  - in (3) we can rewrite the product of s and t as the product of the minimum times the maximum of them, whichever they are.

Example 4.4.2 (Distribution of stopping time of SBM)**.** Let X be a standard brownian motion ($\sigma^2 = 1$). Fix a positive constant $a \in \mathbb{R}$, $a > 0$ and define a stopping time in this way

$$\tau = \inf\{t : X_t = a\}$$

τ is the first time such that the path of the brownian motion reaches the level a (with the usual convention that $\inf \emptyset = +\infty$).
Our goal is to determine the distribution function of τ, $\mathbb{P}(\tau \le t)$.
For $t \le 0$ we have $\mathbb{P}(\tau \le t) = 0$ (starting the brownian motion from $t = 0$. So let $t > 0$; the probability we're interested can be written as splitted in:

$$\begin{aligned}
\mathbb{P}(\tau \le t) &\overset{(1)}{=} \mathbb{P}(\tau \le t, X_t = a) + \mathbb{P}(\tau \le t, X_t < a) + \mathbb{P}(\tau \le t, X_t > a) \\
&\overset{(2)}{=} 0 + \mathbb{P}(\tau \le t, X_t < a) + \mathbb{P}(\tau \le t, X_t > a) \\
&\overset{(3)}{=} 2\,\mathbb{P}(\tau \le t, X_t > a) \\
&\overset{(4)}{=} 2\,\mathbb{P}(X_t > a) \\
&\overset{(5)}{=} 2\,\mathbb{P}\left(\frac{X_t}{\sqrt{t}} > \frac{a}{\sqrt{t}}\right) \\
&\overset{(6)}{=} 2\,\mathbb{P}\left(\mathrm{N}(0,1) > \frac{a}{\sqrt{t}}\right) \\
&= 2\left[1 - \Phi\left(\frac{a}{\sqrt{t}}\right)\right]
\end{aligned}$$

where in

- (1) we split the event in three incompatible events for which we can sum the probability
- (2) we have that

  $$\mathbb{P}(\tau \leq t, X_t = a) \leq \mathbb{P}(X_t = a) = 0$$

  where the first inequality since the event $\{\tau \leq t, X_t = a\} \subseteq \{X_t = a\}$ and furthermore $\mathbb{P}(X_t = a) = 0$ (being X_t normal it's absolutely continuous and therefore the probability of a single point is null).
- (3) it can be shown that (take it as given)

  $$\mathbb{P}(\tau \leq t, X_t < a) = \mathbb{P}(\tau \leq t, X_t > a)$$

- (4), since X has continuous path (up to equivalence, but think an increasing function), with $X_0 = 0$ and $X_t > a$, there must exist a point $s \in [0, t)$, such that $X_s = a$: hence $\tau \leq s < t$ (τ is the first time the path of the brownian motion reach a). In other terms the event $\{X_t > a\} \subset \{\tau \leq t\}$ and thus
- (5) we merely divide both terms by $\sqrt{t}$
- (6) we note that being X_t normal with mean 0 and variance t), then $X_t/\sqrt{t}$ is distributed as standard normal.

To sum up the distribution is the following:

$$\mathbb{P}(\tau \leq t) = 2\left[1 - \Phi\left(\frac{a}{\sqrt{t}}\right)\right]$$

So far $a > 0$ was a positive constant. Suppose now that $a < 0$; in this case

$$\tau = \inf\{t : X_t = a\} = \inf\{t : -X_t = -a\} \stackrel{(1)}{=} \inf\{t : -X_t = |a|\}$$

where in (1) since if $a < 0$, $-a$ is $|a|$. Here

1. $-X$ is still a standard brownian motion (from the definition, is still adapted, take value 0 at 0, has independent increments, and $-X_t$ is normal with mean 0 and variance t). So similarly to the fact that if $Z \sim \mathrm{N}(0, 1)$ then $-Z \sim \mathrm{N}(0, 1)$ here both X and $-X$ are standard brownian motion.
2. with $|a|$ positive

we can repeat the same argument as before, and we end up with an equation valid for all cases

$$\mathbb{P}(\tau \leq t) = 2\left[1 - \Phi\left(\frac{|a|}{\sqrt{t}}\right)\right], \quad \forall t > 0, \forall a \neq 0$$

Here if $a > 0$ $|a|$ doesn't change, but if $a < 0$ absolute value is needed.

Example 4.4.3 (Probability of reaching a in finite time)**.** Byu using result above, let us evaluate the probability of τ being finite (that is the probability that sooner or later the path of the brownian motion reach the level a):

$$\mathbb{P}(\tau < +\infty) = \lim_{t\to+\infty} \mathbb{P}(\tau \leq t) = \lim_{t\to+\infty} 2\left[1 - \Phi\left(\frac{\alpha}{\sqrt{t}}\right)\right] = 2\left[1 - \Phi(0)\right] = 1$$

So almost surely the brownian motion reach the value a (sooner or later). If X describes the price of an action, we fix a level $a \neq 0$ and decide that when it will become a then we sell. By this theorem this will happen, maybe it will take gozilion of years, but will happen

Remark 50*.* Since $\tau < +\infty$, every time we know that something will happen almost surely, a natural question is "how much time si needed to reach teh level a". To answer this question at least roughly let us evaluate $\mathbb{E}[\tau]$, providing this a rough estimate of a time required to reach a.

Example 4.4.4 (Expected time for reaching a)**.** τ is a continuous random variable so expected value we start by obtaining its density, first derivative of the distribution function:

$$\begin{aligned}\frac{\partial}{\partial t}\mathbb{P}(\tau \leq t) &= \frac{\partial}{\partial t}2\left[1 - \Phi\left(\frac{|a|}{\sqrt{t}}\right)\right] = -2\Phi'\left(\frac{|a|}{\sqrt{t}}\right)\cdot|a|\left(-\frac{1}{2}t^{-3/2}\right)\\ &\overset{(1)}{=} \frac{|a|}{t^{3/2}}\cdot\frac{e^{-\frac{1}{2}\frac{a^2}{t}}}{\sqrt{2\pi}}\end{aligned}$$

where in (1), $\Phi'\left(\frac{|a|}{\sqrt{t}}\right)$ is merely the density of a standard normal evaluated in $|a|/\sqrt{t}$. Thus, to evaluate the expected value:

$$\begin{aligned}\mathbb{E}[\tau] &= \int_0^{+\infty} t\cdot f(t)\,\mathrm{d}t = \int_0^{+\infty} t\cdot\frac{|a|}{t^{3/2}}\cdot\frac{e^{-\frac{1}{2}\frac{a^2}{t}}}{\sqrt{2\pi}} = \frac{|a|}{\sqrt{2\pi}}\int_0^{+\infty}\frac{t}{t^{3/2}}e^{-\frac{a^2}{2}\frac{1}{t}}\,\mathrm{d}t\\ &\overset{(1)}{=} +\infty\end{aligned}$$

where in (1) if $t \to +\infty$ the term $e^{-\frac{a^2}{2}\frac{1}{t}} \to 1$ while $\int_0^{+\infty}\frac{1}{\sqrt{t}}\,\mathrm{d}t = +\infty$. So to sum up:

$$\begin{aligned}&\mathbb{P}(\tau < +\infty) = 1\\ &\mathbb{E}[\tau] = +\infty\end{aligned}$$

The interpretation is that, even if sooner or later, the path of X will pick a, the time to reach a is expected to be very large (infact $\mathbb{E}[\tau] = +\infty$). Using the language of Markov chains, state we could say that a is null recurrent state.

Remark 51*.* Next let us discuss a further property of SBM

Proposition 4.4.1. *Let X be a standard brownian motion, fix $t > 0$ and subdivide the interval $[0,t]$ with the following ticks:*

$$0 = t_0^n < t_1^n < t_2^n < \ldots < t_{k_n}^n = t$$

If the maximum length of the sub-intervals goes to 0 as $n \to +\infty$

$$\max_j (t_j^n - t_{j-1}^n) \to 0, \quad \textit{as } n \to \infty$$

then sum of squared differences/increment of process at selected times converge in L_2 to the length of the interval:

$$\sum_{j=1}^{k_n} \left(X_{t_j^n} - X_{t_{j-1}^n} \right)^2 \xrightarrow{L_2} t$$

Example 4.4.5. For instance, let $t = 1$ and subdivide $[0, 1]$ using $t_0^n = 0$ and $t_j^n = \frac{j}{n}$ for $j = 1, \ldots, n$ (that is for each n I split the unit interval into n subintervals of equal length). Then:

$$\max_j \left(t_j^n - t_{j-1}^n \right) = \max_j \frac{1}{n} = \frac{1}{n} \to 0$$

here we can say that

$$\sum_{j=1}^{n} \left(X_{\frac{j}{n}} - X_{\frac{j-1}{n}} \right)^2 \xrightarrow{L_2} 1$$

where 1 is the length of the interval.

TODO: se nn vi è in probabilita mettercelo

Proposition 4.4.2 (Ricordella?)**.** *In general $Y_n \xrightarrow{L_2} c$ with $c \in \mathbb{R}$, if and only if*

$$\mathbb{E}[Y_n] \to c \textit{ and } \mathrm{Var}[Y_n] \to 0$$

Dimostrazione. Infact

$$\begin{aligned}
\mathbb{E}\left[(Y_n - c)^2\right] &\overset{(1)}{=} \mathbb{E}\left[\left((Y_n - \mathbb{E}[Y_n]) + (\mathbb{E}[Y_n] - c)\right)^2\right] \\
&= \underbrace{\mathbb{E}\left[(Y_n - \mathbb{E}[Y_n])^2\right]}_{\mathrm{Var}[Yn]} + \underbrace{\mathbb{E}\left[(\mathbb{E}[Y_n] - c)^2\right]}_{(\mathbb{E}[Y_n]-c)^2} + \mathbb{E}\left[2(Y_n - \mathbb{E}[Y_n])(\mathbb{E}[Y_n] - c)\right] \\
&= \mathrm{Var}[Y_n] + (\mathbb{E}[Y_n] - c)^2 + 2(\mathbb{E}[Y_n] - c) \underbrace{\mathbb{E}[Y_n - \mathbb{E}[Y_n]]}_{=0} \\
&= \mathrm{Var}[Y_n] + (\mathbb{E}[Y_n] - c)^2
\end{aligned}$$

Therefore $\mathbb{E}\left[(Y_n - c)^2\right] \to 0 \iff \mathbb{E}[Y_n] \to c$ and $\mathrm{Var}[Y_n] \to 0$.

□

Dimostrazione. Now to prove the theorem, we let

$$Y_n = \sum_{j=1}^{k_n} \left(X_{t_j^n} - X_{t_{j-1}^n} \right)^2$$

Then it suffices to prove that $\mathbb{E}[Y_n] \to t$ and $\mathrm{Var}[Y_n] \to 0$. So:

- regarding the expected value

$$\begin{aligned}
\mathbb{E}\left[Y_n\right] &= \mathbb{E}\left[\sum_{j=1}^{k_n}\left(X_{t_j^n}-X_{t_{j-1}^n}\right)^2\right] = \sum_{j=1}^{k_n}\mathbb{E}\left[\left(X_{t_j^n}-X_{t_{j-1}^n}\right)^2\right] \\
&\overset{(1)}{=} \sum_{j=1}^{n}\left(t_j^n - t_{j-1}^n\right) \\
&= t_1^n - t_0^n + t_2^n - t_1^n + \dots t_{k_n}^n - t_{k_{n-1}}^n \\
&= t_{k_n}^n - t_0^n = t - 0 \\
&= t
\end{aligned}$$

where in (1) since X is a standard brownian motion then $X_t - X_s \sim \mathrm{N}(0, t-s)$, and since its the sum of variance of the increments (which are independent) its the sum of the variances.
So being even equal to t is goes to t as well

- regarding variance

$$\begin{aligned}
\mathrm{Var}\left[Y_n\right] &= \mathrm{Var}\left[\sum_j\left(X_{t_j^n}-X_{t_{j-1}^n}\right)^2\right] \overset{(1)}{=} \sum_j \mathrm{Var}\left[\left(X_{t_j^n}-X_{t_{j-1}^n}\right)^2\right] \\
&\overset{(2)}{\leq} \sum_j \mathbb{E}\left[\left(X_{t_j^n}-X_{t_{j-1}^n}\right)^4\right] \overset{(3)}{=} \sum_j \mathbb{E}\left[\left(\frac{X_{t_j^n}-X_{t_{j-1}^n}}{\sqrt{t_j^n - t_{j-1}^n}}\right)^4 \left(t_j^n - t_{j-1}^n\right)^2\right] \\
&\overset{(4)}{=} \sum_j \left(t_j^n - t_{j-1}^n\right)^2 \underbrace{\mathbb{E}\left[\mathrm{N}(0,1)^4\right]}_{\text{it's 3 btw}} \\
&\overset{(5)}{\leq} \mathbb{E}\left[\mathrm{N}(0,1)^4\right] \cdot \max_j\left(t_j^n - t_{j-1}^n\right) \cdot \underbrace{\sum_j\left(t_j^n - t_{j-1}^n\right)}_{=t} \\
&= t \cdot \mathbb{E}\left[\mathrm{N}(0,1)^4\right] \max_j\left(t_j^n - t_{j-1}^n\right)
\end{aligned}$$

where in:

  - (1) by independence of the increments (and the variance of the sum of independent random variables has no covariance)
  - (2) since that in general for any rv Y we have that $\mathrm{Var}[Y] \leq \mathbb{E}\left[Y^2\right]$
  - (3) we divided and multiplied the difference of the two variables for their standard deviation in order to standardize
  - (4) we have that $\frac{X_{t_j^n}-X_{t_{j-1}^n}}{\sqrt{t_j^n - t_{j-1}^n}}$ is standard normal because is the difference of the variable with his mean (0) and divided by the standard deviation
  - (5) to construct the upper bound of the sum of squares square we choosed the maximum value multiplied for the remaining

So looking at the final equation the term $\max_j\left(t_j^n - t_{j-1}^n\right)$ by assumption goes to 0 as $n \to +\infty$. Therefore $\mathrm{Var}[Y_n] \to 0$ as $n \to +\infty$.

□

4.5 BM and strong law of large numbers

Remark 52. An important property of BM is that it satisfies a strong law of large number.

Theorem 4.5.1. *A brownian motion X satisifies the strong law of large numbers in the sense that*

$$\frac{X_t}{t} \xrightarrow{a.s.} 0, \quad \text{for } t \to +\infty$$

Dimostrazione. We prove this not in general, but in a special case where the $\lim tto + \infty$ is taken along integers $\mathbb{N}$ and not the reals (namely we consider $\frac{X_n}{n}$ instead of $\frac{X_t}{t}$).
We have that X_n/n can be written as

$$\frac{X_n}{n} \stackrel{(1)}{=} \frac{X_n - X_0}{n} \stackrel{(2)}{=} \frac{\sum_{i=1}^{n}(X_i - X_{i-1})}{n}$$

where:

- in (1) since $X_0 = 0$ almost surely in BM
- in (2) to see it, we have that

$$\sum_{i=1}^{n}(X_i - X_{i-1}) = X_1 - X_0 + X_2 - X_1 + \ldots + X_n - X_{n-1} = X_n - X_0$$

But since $X_i - X_{i-1} \sim \mathrm{N}\left(0, \sigma^2 \cdot 1\right)$ are iid (because BM stationary independent increment) then we can apply kolmogorov strong law of large number. Thus

$$\frac{\sum_{i=1}^{n}(X_i - X_{i-1})}{n} \xrightarrow{a.s.} \underbrace{\mathbb{E}\left[X_1 - X_0\right]}_{\text{common mean of } X_i - X_{i-1}} = \mathbb{E}\left[X_1\right] = 0$$

□

Capitolo 5

Poisson and related processes

5.1 Introduction

Definition 5.1.1 (Poisson process)**.** A process $N = \{N_t : t \geq 0\}$ indexed by non negative reals (eg $t \in T = [0, +\infty)$) is a Poisson process if:

1. N is adapted to some filtration $(\mathscr{F}_t)$;

2. N has independent increments;

3. $N_0 = 0$ almost surely and $N_t \sim \text{Pois}(\lambda t)$, $\forall t > 0$ where $\lambda > 0$ is a parameter

Remark 53. Poisson process are quite similar to BM:

1. looking at definition, the only difference for N poisson process is that $N_t \sim \text{Pois}(\lambda \cdot t)$, $\forall t > 0$, while for brownian motion X we have $X_t \sim \text{N}\left(0, \sigma^2 t\right)$, $\forall t > 0$.

2. it holds a similar characterization depending on features of the path they produces, the sense that if X is a process such that $X_0 = 0$ almost surely, then

   X is BM (wrt its canonical filtration)
   $\iff$
   X has stationary independent increments and continuous path up to equivalence

   similarly

   X is Poisson process (wrt its canonical filtration)
   $\iff$
   X has stationary independent increments and has path which are
   right continuous, non decreasing, piecewise constant, with unitary jumps
   and having countably many jumps on all of $[0, \infty)$.

   So considering this characterization the main (and in a sense the only difference) is that path are continuous with BM or satisfy condition (*)

with poisson process.
To remind all the properties of the path think we're counting random arrivals[1]: thus, the *shape of the path* is a *starway* starting from time 0 with value 0. The process remains 0 until the first arrival; then is right continuous with unitary increases that occurs at time where an arrival occurs. Jumps are unitary because we assume there are no events that occurs in the same moment.

Proposition 5.1.1 (Stationary increments and deltas distribution)**.** *A Poisson process has not only independent increment (by the definition) also* stationary increments*, that is*

$$N_t - N_s \sim N_{t-s} - N_0$$

and since in case of poisson $N_0 = 0$ *we can say*

$$N_t - N_s \sim N_{t-s} - N_0 = N_{t-s} \sim \mathrm{Pois}\left(\lambda(t-s)\right)$$

Dimostrazione. To prove $N_{t-s} \sim \mathrm{Pois}\left(\lambda(t-s)\right)$ it suffices to argue as in case of BM. Namely $N_t = N_s + (N_t - N_s)$ and $N_s \perp\!\!\!\perp (N_t - N_s)$ by independence of the increments. Hence using characteristic functions on obtains

$$\varphi_{N_t}(u) = \varphi_{N_s}(u) \cdot \varphi_{N_t - N_s}(u)$$

Or equivalently

$$\varphi_{N_t - N_s}(u) = \frac{\varphi_{N_t}(u)}{\varphi_{N_s}(u)} = \frac{\text{char fun of Pois}\left(\lambda t\right)}{\text{char fun of Pois}\left(\lambda s\right)} = \text{char fun of Pois}\left(\lambda(t-s)\right)$$

□

5.2 Stopping times and counting process

Remark 54. In order to better understand that a Poisson process is actually a counting process, let's introduce the stopping times

Definition 5.2.1. In the Poisson process the stopping times are defined as

$$\begin{aligned}
&\tau_0 = 0\\
&\tau_1 = \inf\{t : N_t = 1\}\\
&\dots\\
&\tau_k = \inf\{t : N_t = k\}
\end{aligned}$$

τ_1 is the time of first arrival, τ_k is first time where we have k arrival. So, on the time line we have that $0 < \tau_1 < \tau_2 < \dots$.

Proposition 5.2.1. *If* N *is a poisson process the sequence* $(\tau_n - \tau_{n-1} : n \geq 1)$ *is iid with common distribution* $\tau_n - \tau_{n-1} \sim \mathrm{Exp}\left(\lambda\right)$

[1] eg at the bus stop you remain for an hour and count the number of cars that passes, or counting the visit to a website in a time interval

Dimostrazione. Here we prove only that $\tau_1 - \tau_0 = \tau_1 \sim \text{Exp}(\lambda)$. [2] Indeed

$$\mathbb{P}(\tau_1 > t) = \mathbb{P}(N_t = 0) = e^{-\lambda t}, \quad \forall t > 0$$

Equivalently

$$\mathbb{P}(\tau_1 \leq t) = 1 - e^{-\lambda t} \forall t > 0$$

and this latter is the distribution function for $\text{Exp}(\lambda)$ □

Remark 55. So the times

- between two arrivals are independents and have same distribution which is an exponential with parameter λ.
- τ are *not* independent (they're increasing, can't be independent) but the deltas $(\tau_n - \tau_{n-1})$ are so $(\tau_1 - 0)$ is independent from $(\tau_2 - \tau_1)$ etc.

Definition 5.2.2 (Poisson process as counting process)**.** Using the stopping times, the process N can be written as follows: Note that

$$N_t = \begin{cases} 0 & \text{if } 0 \leq t < \tau_1 \\ 1 & \text{if } \tau_1 \leq t < \tau_2 \\ \dots & \dots \end{cases}$$

or equivalently

$$N_t = j \iff t \in [\tau_j, \tau_{j+1}), \quad \forall j = 0, 1, 2, \dots$$

so that N_t can be regarded as

$$N_t = \{\text{number of arrivals in the interval } [0, t]\}$$

Example 5.2.1. Given $0 < s < t$ and $j = 0, 1, \dots, n$, we want to evaluate the probability

$$\mathbb{P}(N_s = j | N_t = n) = \mathbb{P}(j \text{ arrivals in } [0, s] \mid n \text{ arrivals in } [0, t])$$

[2] Dimostrazione fatta il mio anno We dont' prove time deltas are independent o identically distributed (take it as given) but we show that the common distribution of $\tau_n - \tau_{n-1}$ is actually $\text{Exp}(\lambda)$.
We start from the complement to 1 of the distribution function:

$$\mathbb{P}(\tau_1 - \tau_0 > t) \overset{(1)}{=} \mathbb{P}(\tau_1 > t) \overset{(2)}{=} \mathbb{P}(N_t = 0) \overset{(3)}{=} e^{-\lambda t} \quad \forall t \geq 0$$

where in

- (1) since $\tau_0 = 0$ by construction
- (2) the event is that at time t we have no arrival (since the first occurs at τ_1 which is following t)
- (3) is the probability that a poisson with λt parameter to take value 0. Remember that if $X \sim \text{Pois}(\lambda)$ then $\mathbb{P}(X = x) = e^{-\lambda} \cdot \frac{\lambda^x}{x!}$ $\forall x \in \{0, 1, \dots\}$, so $\mathbb{P}(X = 0) = e^{-\lambda}$

So with $e^{-\lambda t}$ we have the complement to 1 of the distribution function of an exponential (which is $1 - e^{-\lambda t}$). Therefore since we started from the complement to 1 of the distribution function of the differences between times we can say that

$$\tau_1 - \tau_0 \sim \text{Exp}(\lambda)$$

This probability

$$
\begin{aligned}
\mathbb{P}\left(N_s = j | N_t = n\right) &\overset{(1)}{=} \frac{\mathbb{P}\left(N_s = j, N_t = n\right)}{\mathbb{P}\left(N_t = n\right)} \\
&\overset{(2)}{=} \frac{\mathbb{P}\left(N_s = j, N_t - N_s = n - j\right)}{\mathbb{P}\left(N_t = n\right)} \\
&\overset{(3)}{=} \frac{\mathbb{P}\left(N_s = j\right) \cdot \mathbb{P}\left(N_t - N_s = n - j\right)}{\mathbb{P}\left(N_t = n\right)} \\
&\overset{(4)}{=} \frac{\dfrac{e^{-\lambda s}(\lambda s)^j}{j!} \dfrac{e^{-\lambda(t-s)}(\lambda(t-s))^{n-j}}{(n-j)!}}{\dfrac{e^{-\lambda t}(\lambda t)^n}{n!}} \\
&= \binom{n}{j} \frac{s^j \cdot (t-s)^{n-j}}{t^n} = \binom{n}{j} \left(\frac{s}{t}\right)^j \left(1 - \frac{s}{t}\right)^{n-j} \\
&= \mathbb{P}\left(\mathrm{Bin}\left(n, \frac{s}{t}\right) = j\right)
\end{aligned}
$$

where the last equation is a binomial with parameter $\frac{s}{t}$ and in

- (1) we apply the definition being $\mathbb{P}\left(N_t = n\right) > 0$;
- (2) we rewrite the second event;
- (3) by independent increments the two random variables at the numerator are independent;
- (4) all are poisson with different parameters due to time indexes.

Remark 56 (Interpretation). Under the information that we have a total of n arrivals up to t, the probability of having j arrivals up to s (with $s < t$ and $j \leq n$) is binomial with parameter $\frac{s}{t}$ In practice this means that suppose we know that $N_t = n$; then conditionally on this information we can say that the arrival times $(\tau_1, \ldots, \tau_n)$ are iid and uniformly distributed on $(0, t)$.

Example 5.2.2. Suppose the value at time t is $n = 1$; we have two possibilities regarding the value at time s

$$
\begin{aligned}
\mathbb{P}\left(N_s = 1 | N_t = 1\right) &= \mathbb{P}\left(\tau_1 \leq s | N_t = 1\right) = \binom{1}{1}\left(\frac{s}{t}\right)^1\left(1 - \frac{s}{t}\right)^0 = 1 \cdot \frac{s}{t} \cdot 1 \\
&= \frac{s}{t} \\
\mathbb{P}\left(N_s = 0 | N_t = 1\right) &= \mathbb{P}\left(\tau_1 > s | N_t = 1\right) = \binom{1}{0}\left(\frac{s}{t}\right)^0\left(1 - \frac{s}{t}\right)^1 = 1 \cdot 1 \cdot \left(1 - \frac{s}{t}\right) \\
&= 1 - \frac{s}{t}
\end{aligned}
$$

So it basically depends on the ratio between, that is it's uniform, and precisely $\mathrm{Unif}\,(0, t)$.

If I know that there is 1 arrival up to t so it happened in interval $(0, t)$: so conditional distribution $\tau_1 | N_t = 1 \sim \mathrm{Unif}\,(0, t)$. On the other hand, however, note that unconditional distribution $\tau_1 \sim \mathrm{Exp}\,(\lambda)$.

$$
\begin{cases}
\tau_1 \sim \mathrm{Exp}\,(\lambda) & \text{unconditionally} \\
\tau_1 | N_t = 1 \sim \mathrm{Unif}\,(0, t) & \text{conditionally}
\end{cases}
$$

Same interpretation holds if we have n arrivals from 0 to t, they are conditionally Unif $(0, t)$.

5.3 Compensated and spatial poisson process

Remark 57. Now two remarks regarding the poisson process

Important remark 28. The poisson process is not a martingale (incidentally this is a further difference with BMS). Infact

$$\mathbb{E}[N_t] = \lambda t \neq 0 = \lambda 0 = \mathbb{E}[N_0]$$

If the poisson process would be a martingale the expectation would be constant in time.
However is easy to *modify* the process to make it a martingale (being the mean not constant we only subtract it to make it constant)

Proposition 5.3.1 (Compensated Poisson process is a martingale)**.** *Let N_t be a Poisson process; the process $\widetilde{N}_t$ defined by subtracting from the original process the expected value $\mathbb{E}[N_t]$:*

$$\widetilde{N}_t = N_t - \lambda t$$

is called compensated poisson process *and is a martingale.*

Dimostrazione. Given $0 \leq s < t$

$$\begin{aligned}
\mathbb{E}\left[\widetilde{N}_t | \mathscr{F}_s\right] &= \mathbb{E}[N_t - \lambda t | \mathscr{F}_s] = -\lambda t + \mathbb{E}[N_t | \mathscr{F}_s] = -\lambda t + \mathbb{E}[N_s + N_t - N_s | \mathscr{F}_s] \\
&\overset{(1)}{=} -\lambda t + N_s + \mathbb{E}[N_t - N_s | \mathscr{F}_s] \overset{(2)}{=} -\lambda t + N_s + \mathbb{E}[N_t - N_s] \\
&\overset{(3)}{=} -\lambda t + N_s + \lambda(t - s) = N_s - \lambda s \\
&= \widetilde{N}_s
\end{aligned}$$

where in

- (1) being N_s being $\mathscr{F}_s$ measurable we can take it out of expectation
- (2) having independent increments we can drop the conditioning
- (3) $N_t - N_s \sim \text{Pois}(\lambda(t - s))$

□

Remark 58. The next remark is more important; it's a modification of the Poisson process which is useful in applications.

Definition 5.3.1 (Spatial Poisson process)**.** Let be $H \subset \mathbb{R}^n$ be a bounded borel subset of $\mathbb{R}^n$; let $\mathscr{U}$ be a collection of borel subset of H and consider a process. To each $A \in \mathscr{U}$ we associate a random variable $N(A)$ and we focus on the process

$$N = \{N(A) : A \in \mathscr{U}\}$$

Then N is said to be a spatial Poisson process if satisfies two properties:

1. if $A_1, \ldots, A_k \in \mathscr{U}$ are pairwise disjoint ($A_i \cap A_j = \emptyset$, $\forall i \neq j$), then the corrisponding random variables $N(A_1), \ldots, N(A_k)$ are independent
2. $N(A) \sim \text{Pois}\,(\lambda \cdot m(A)), \forall A \in \mathscr{U}$, where m is Lebesgue measure on $\mathbb{R}^n$. Eg in dimension $n = 1$ the meaning of lebesgue measure is the length, in dimension $n = 2$ the area, in dimension 3 the volume and so on; if we are working in dim 2 the random variable depends on a lambda and something connected to the area of A

Example 5.3.1. Associamo ad ogni elemento A di $\mathscr{U}$ una variabile aleatoria: for instance $H = \text{ITALY}$, $\mathscr{U} = \{\text{italian regions}\}$, and

$$N(A) = \{\text{number of cases of COVID in a given time interval in region A}\}$$

If we want to use a poisson process in the italian region we're making the two assumptions above. Are these reasonable? no they're not reasonable, especially the first: the number of cases in emilia is not independent of the number of disease in tuscany. (it is not reasonable $N(A) \perp\!\!\!\perp N(B)$ even if $A \cap B = \emptyset$)

5.4 Levy processes

Definition 5.4.1 (Levy process)**.** A process $X = \{X_t : t \geq 0\}$ indexed by $[0, +\infty)$ is said to be a Levy process with respect to a filtration $(\mathscr{F}_t)$, if

1. X is adapted to $(\mathscr{F}_t)$
2. $X_0 = 0$ almost surely
3. X has stationary indipendent increments
4. $X_s \xrightarrow{p} X_t$ if $s \to t$

Remark 59. The two main examples of Levy process are Brownian motion and Poisson process. The first three condition are by definition; we want to prove the fourth holds for both the processes

Dimostrazione. Regarding:

- BM, fourth property is true because the paths of BM are continuous up to equivalence; if $s \to t$, $X_s \xrightarrow{p} X_t$.
- poisson process, what does $X_s \xrightarrow{p} X_t$ mean? Given $\varepsilon > 0$ we have to prove

  $$\lim_{s \to t} \mathbb{P}\left(|X_s - X_t| > \varepsilon\right) = 0, \quad \forall \varepsilon > 0$$

  infact, assuming for simplicity $s < t$, one obtains that the distribution of $X_s - X_t$ is Poisson and $\forall \varepsilon \in ()$

  $$\begin{aligned} \mathbb{P}\left(|X_s - X_t| > \varepsilon\right) = \mathbb{P}\left(\text{Pois}\left(\lambda(t-s)\right) > \varepsilon\right) &\overset{(1)}{=} \mathbb{P}\left(\text{Pois}\left(\lambda(t-s)\right) > 0\right) \\ &= 1 - \mathbb{P}\left(\text{Pois}\left(\lambda(t-s)\right) = 0\right) \\ &= 1 - e^{-\lambda(t-s)} \end{aligned}$$

where in (1) supposing that the small ε is $0 < \varepsilon < 1$ the only value which does not satisfies $X_{|t-s|} > \varepsilon$ is 0.
So if $s \to t$ then $1 - e^{-\lambda(t-s)} \to 1 - e^0 = 0$ and this proves the fouth condition of the levy process for a poisson one.

□

Remark 60. Another interesting Levy process is the following.

Definition 5.4.2 (Compound Poisson process). It's defined as

$$X_t = \mathbb{1}(N_t > 0) \cdot \sum_{j=1}^{N_t} Z_j, \quad \forall t \geq 0$$

where N is a poisson process, (Z_j) is iid and (Z_j) independent of N

Important remark 29 (Interpretation). What is the idea of the process:

- we start with a Poisson process N and we have the arrival times of a poisson process $0 < \tau_1 < \tau_2 < \dots$ (with τ_1 time of first arrival);
- if in the Poisson process jumps are unitary, here are iid random variables Z_j (of any kind): so the graph with the path which start at 0, at τ_1 will goes to Z_1, at τ_2 will goes to $Z_1 + Z_2$ (which differently from poisson is not needed to be $Z_1 + 1$, can be lower than Z_1 if Z_2 is negative) and so on.

clearly if $Z_j = 1 \; \forall j$ we obtain the Poisson process: in other words, teh Poisson process is a special case of the compound Poisson process.

5.5 Renewal processes

Remark 61. It's a generalization of Poisson process.

Definition 5.5.1 (Renewal process). Let (Z_n) be a sequence of independent random variables such that

- $Z_n > 0$ almost surely $\forall n \geq 1$
- $Z_2 \sim Z_n \forall n \geq 2$ (the sequence starting from the *second* variable on is iid; it may be that all (Z_n) is iid if even $Z_1 \sim Z_n$ or not)

Given the sequence (Z_n) define

- $\tau_0 = 0$, $\tau_n = \sum_{i=1}^{n} Z_i$. Since $Z_n > 0, \forall n \geq 1$ we have $0 = \tau_0 < \tau_1 < \tau_2 < \dots$, can be represented on the time line and interpretation is the same of the Poisson process: τ_1 is time of first arrival/renewal, τ_2 of the second and son on;
- the following process is called renewal process

$$N_t = \max \{n \geq 0 : \tau_n \leq t\}, \forall t \in [0, +\infty)$$

Thus, once again $N = \{N_t : t \geq 0\}$ is a process indexed by $[0, +\infty)$. Furthermore

$$\begin{cases} N_t = 0 & \text{if } t \in [0, \tau_1) \\ N_t = 1 & \text{if } t \in [\tau_1, \tau_2) \\ \dots \end{cases}$$

And in general

$$N_t = n \iff \tau_n \leq t < \tau_{n+1}$$

Important remark 30. some remarks:

- while (Z_n) is a *discrete* sequence, N_t is a process in continuous time, and can be regarded as the number of arrivals (or *renewals*, in this process language) in the interval $[0, t]$.
  So it's a counting process just as the Poisson process. However
  - in the special case of Poisson process the sequence $(\tau_n - \tau_{n-1})_{n\geq 1}$) is iid $\text{Exp}(\lambda)$
  - here $(\tau_n - \tau_{n-1})_{n\geq 1}$ is iid with an arbitrary distribution and it may be that $(\tau_1 - \tau_0) = \tau_1 \nsim (\tau_2 - \tau_1)$
- regarding the first two random variables Z_1, Z_2, both the following possibilities are allowed by the general definition above: $Z_1 \sim Z_2$ or $Z_1 \nsim Z_2$. Why Z_2 is not forced to have the same distribution of Z_1 in the general framework?
  Note that Z_1 is time of first arrival, while Z_2 is the time between the first and the second (and in general $Z_n = \tau_n - \tau_{n-1}$, for $n \geq 2$ is the time between the arrivals $(n-1)$ and n).
  In real processes/applications time of first arrival doesn't have to be the same distribution of the time between the first and second arrival (eg first and second girl/boyfriend $Z_2 \nsim Z_1$).
  The Poisson process is a special case that is obtained assuming further that $Z_1 \sim Z_2 \sim \text{Exp}(\lambda)$ (while in the general case above $Z_1 \sim Z_2$ is not required).
- the Poisson process is the *only* renewal process such that $Z_2 \sim Z_1 \sim \text{Exp}(\lambda)$

Remark 62. A useful result concerning renewal processes is teh following SLLN

Theorem 5.5.1. *If N is a renewal process*

$$\frac{N_t}{t} \xrightarrow{a.s.} \begin{cases} 0 & \text{if } \mathbb{E}[Z_2] = +\infty \\ \frac{1}{\mathbb{E}[Z_2]} & \text{if } \mathbb{E}[Z_2] < +\infty \end{cases} \quad \text{for } t \to \infty$$

Remark 63. This result is an easy consequence of the classical Kolmogorov's SLLN. Applying it to the special case of poisson process (ie $Z_1 \sim Z_2 \sim \text{Exp}(\lambda)$) one obtains

$$\frac{N_t}{t} \xrightarrow{a.s.} \frac{1}{\mathbb{E}[Z_2]} = \frac{1}{1/\lambda} = \lambda, \quad \text{as } t \to \infty$$

This fact provides an interpretation of the parameter λ of a Poisson process. Indeed $\lambda \overset{(a.s)}{=} \lim_{t\to+\infty} \frac{N_t}{t}$ so that λ can be regarded as the limit intensity of the random arrivals. Recall that $N_t =$"n of arrivals in $[0,t]$" so that N_t/t can be thought of as the intensity of arrivals in $[0,t]$. Roughly speeaking it is the number of arrivals over the length of the interval.

Theorem 5.5.2. *If* $\mathbb{E}[Z_2] < +\infty$*, then*

NB: da qui in fondo dell'anno scorso, eventualmente integra

$$\frac{N_t}{t} \xrightarrow{a.s.} \frac{1}{\mathbb{E}[Z_2]},$$

Dimostrazione. In fact by definition if $N_t = n$ with $t \in [\tau_{N_t}, \tau_{N_t+1})$ that is $\tau_{N_t} \le t < \tau_{N_t+1}$. Hence

$$\frac{N_t}{\tau_{N_t+1}} < \frac{N_t}{t} \le \frac{N_t}{\tau_{N_t}}$$

Now looking separately at the two margin for $t \to \infty$:

- we have that

$$\frac{N_t}{\tau_{N_t}} = \frac{1}{\frac{\tau_{N_t}}{N_t}} \overset{(1)}{=} \frac{1}{\frac{\sum_{i=1}^{N_t} Z_i}{N_t}} \overset{(2)}{=} \frac{1}{\mathbb{E}[Z_2]}$$

  in

  - (1) by definition of τs
  - (2) rv starting from the second are iid so we use kolmogorov and the denominator goes to the common mean which is the expectation of Z_2 as $t \to +\infty$ (index of the sum $N_t \xrightarrow{a.s.} +\infty$ as $t \to +\infty$)

- similarly

$$\frac{N_t}{\tau_{N_t+1}} \overset{(1)}{=} \frac{N_t}{N_t+1} \cdot \frac{1}{\frac{\tau_{N_t+1}}{N_t+1}} \overset{(2)}{\to} 1 \cdot \frac{1}{\mathbb{E}[Z_2]}$$

  where in

  - (1) we merely multiplied and divided by $N_t + 1$ at denominator
  - (2) since $N_t \to +\infty$ as $t \to +\infty$, then $\frac{N_t}{N_t+1} \xrightarrow{a.s.} 1$ as $t \to \infty$; for the other term the same as before

Thus being $\frac{N_t}{t}$ bounded by terms which goes to $1/\mathbb{E}[Z_2]$ as $t \to +\infty$, it behaves the same way. □

Remark 64. This theorem applies to any renewal process. In the following the application to the Poisson process.

Example 5.5.1. Applying this thm to the Poisson process one obtains that:

$$\frac{N_t}{t} \xrightarrow{a.s.} \frac{1}{\mathbb{E}[Z_2]} \overset{(1)}{=} \lambda$$

where in (1) we have that if $Z_2 \sim \text{Exp}(\lambda)$, then has mean $1/\lambda$.
So in the special case of the Poisson process N_t/t goes almost surely to λ. This fact helps to understand the meaning of parameter λ of Poisson process; being

$$\frac{N_t}{t} = \frac{\text{number of arrivals in } [0,t]}{t}$$

λ is a sort of asymptotic "intensity" of the process as $t \to +\infty$ (number of arrivals over legth of the interval).

Capitolo 6

Other topics

6.1 Inequalities for (sub-)martingales

Remark 65. In the martingale framework, some inequalities are available

Proposition 6.1.1. *If* (X_n) *is a submartingale*

$$\mathbb{P}\left(\max_{0\leq j\leq n} X_j > c\right) = \frac{1}{c}\,\mathbb{E}\left[X_n^+\right], \quad \forall c > 0$$

where $X_n^+ = \max\{X_n, 0\}$

Example 6.1.1. As application, suppose $X_0 = 0$ and $X_n = \sum_{i=1}^n Z_i$, where (Z_i) is iid with $\mathbb{E}[Z_1] = 0$, $\mathbb{E}\left[Z_1^2\right] = 1$. Then (X_n) is a martingale and (X_n^2) is a submartingale. Hence applying what above:

$$\begin{aligned}\mathbb{P}\left(\max_{0\leq j\leq n} |X_j| > c\right) &= \mathbb{P}\left(\max_{0\leq j\leq n} X_j^2 > c^2\right) \\ &\leq \frac{1}{c^2}\,\mathbb{E}\left[X_n^2\right] = \frac{n}{c^2}\,\mathbb{E}\left[Z_1^2\right] \\ &= \frac{n}{c^2}\end{aligned}$$

where $\mathbb{E}\left[X_n^2\right] = n$ because it's the sum of rvs with variance 1.
Note that, in this case, Tchebychev inequality only yields

$$\mathbb{P}\left(\max_{0\leq j\leq n} |X_j| > c\right) \leq \frac{1}{c^2}\,\mathbb{E}\left[\max_{0\leq j\leq n} X_j^2\right]$$

In other terms, in this case, the upper bound provided by Tchebychev is worse.

Proposition 6.1.2. *If* (X_n) *is a submartingale and* $X_n \geq 0, \forall n$*, then*

$$\mathbb{E}\left[\max_{0\leq j\leq n} X_j^p\right] \leq \left(\frac{p}{p-1}\right)^p \mathbb{E}\left[X_n^p\right], \quad \forall p > 1$$

Example 6.1.2. For instance if $p = 2$

$$\mathbb{E}\left[\max_{0\leq j\leq n} X_j^2\right] \leq \left(\frac{2}{2-1}\right)^2 \mathbb{E}\left[X_n^2\right] = 4\,\mathbb{E}\left[X_n\right]$$

6.2 Continuous time martingale

Remark 66. Recall that a process $X = (X_t : t \geq 0)$, indexed by $T = [0, +\infty)$ is a martingale, with respecto to a filtration $(\mathscr{F}_t)$, if

- X is adapted to $(\mathscr{F}_t)$
- $\mathbb{E}[|X_t|] < +\infty, \forall t$
- $\mathbb{E}[X_t | \mathscr{F}_t] = X_s$, $\forall 0 \leq s < t$

So far we discussed mainly discrete time martingales. The only things we know regarding continuous time martingales are that: BM is a martingale while Poisson process is not.
Now we mention a result concerning continuous time martingales, which essentially states that the Doob-Meyer decomposition (already discussed in discrete time) can be extended to continuous martingale.

Theorem 6.2.1. *If X is a submartingale $X_t \geq 0, \forall t$, and X has continuous paths, then X can be written as*

$$X_t = M_t + A_t, \quad \forall t \geq 0$$

where M is a martingale with continuous paths and A is an adapted process with continuous and increasing paths

6.3 Finite dimensional distributions

Definition 6.3.1. Let $X = (X_t : t \in T)$ be any process indexed by an arbitrary set T. The finite dimensional distributions (FDD) of X are merely the probability distributions of the random vectors $(X_{t_1}, X_{t_2}, \ldots, X_{t_n})$, $\forall n \geq 1$, $\forall t_1, \ldots t_n \in T$.

Remark 67. In other terms the FDD of X are nothing but the distributions of the all the random vectors which can be obtained by specifying a finite numbers of point of T

Remark 68. In real problems, when one aims to describe a real phenomenon by a process X, so that one has to select X, on first decides the FDD of X. In other terms, one argue as follows: “in order to model the phenomenon, it would be desirable that X_t has some distribution, $\forall t$, (X_s, X_t) has some distribution $\forall (s,t)$, (X_s, X_t, X_u) has some distribution $\forall (s,t,u)$, and so on.
In other terms, in order to select a process X we first choose the FDD that such X should have.
But here a problem arises: the FDD cannot be arbitrary but they must satisfy some conditions, usually called the **consistency condition**, in order for a process X having such FDD actually exists.
Suppose that, in order to model some phenomenon I want $X_t \sim \text{Bin}, \forall t$ and $(X_s, X_t) \sim \text{N}$. This is impossible! In fact if $(X_1, X_t) \sim \text{MVN}$ then $X_s \sim \text{N}$ and $X_t \sim \text{N}$ (so that X_s and X_t are not binomial)

Remark 69. To sum up, when we choose the FDD that a process X should have, we are not totally free since the FDD must satisfy some conditions. Luckily there are some thms which provide conditions on the FDD which imply that a process X having such FDD actually exists. The most popular is the so called *Kolmogorov Consistency theorem*. Another analogous is the *Ionescu-Tulcea theorem*. In this course we don't state these thms formally; for our purposes, it suffices to know that such thms are actually available. A final example

Example 6.3.1. Suppose that to describe some phenomenon, we need a discrete time process $X = (X_n : n \geq 0)$ (in other terms $T = \{0, 1, 2, \ldots\}$). Suppose also that

- we choose the distribution of X_0 arbitrarily
- $\forall x$ we choose the conditional distribution of X_1, given $X_0 = x$, $\mathbb{P}(X_1 | X_0 = x)$ arbitrarily
- then we go on according to this rule: that is $\forall x_0, x_1, \ldots, x_n$ we chose the conditional distribution of X_{n+1} given $X_0 = x_0, X_1 = x_1, \ldots, X_n = x_n$ arbitrarily

Then, thanks to the Ionescu-Tulcea thm, there *exists* a process $X = (X_n : n \geq 0)$ having the prescribed conditional distributions.